



KTU NOTES

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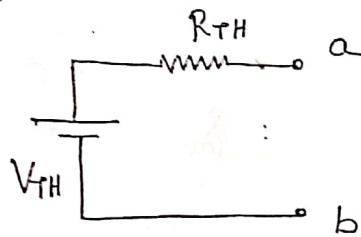
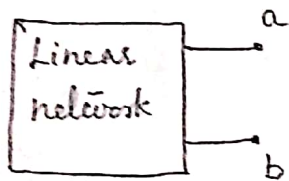
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Network Theorems - Module 1

In complicated networks with a number of voltage sources and resistances, application of Ohm's law and Kirchhoff's laws will not yield a solution. To simplify such circuits and to get an easy solution to that, a number of network theorems are developed. These theorems are applicable both to dc and ac circuits. In ac circuits, impedance replaces resistance of dc system.

Thevenin's theorem - states that any linear network of resistances and voltage sources if viewed from any two points 'a' & 'b' in the network can be replaced by a single source of emf and a single resistance in series with the source of emf.

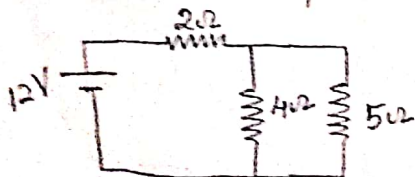
The single source of emf is the open circuit voltage across the terminals (when resistance connected across that terminal is removed) and the single resistance is the internal resistance of the network with all voltage sources short-circuited (or all voltage sources are replaced by their internal resistance, if the sources have internal resistance) and current sources open-circuited.



Thevenin's equivalent circuit

Explanation

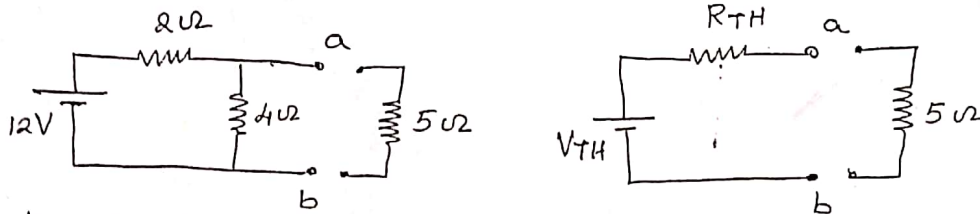
Consider a simple circuit shown here



Find the current flow through 5Ω resistance.

Solution
It is required to calculate the current flow through 5Ω .

To calculate the current flow through 5Ω by applying Thevenin's theorem, disconnect 5Ω resistor from the network and obtain the Thevenin's equivalent circuit about those terminals (a & b) from which 5Ω is disconnected



Thus to start with, disconnect 5Ω

To find V_{TH} (Thevenin's equivalent voltage)



V_{TH} is the open circuit voltage across the terminals a & b

$$\text{i.e. } V_{TH} = V_{ab} = \text{Voltage across } 4\Omega \\ = I_{4\Omega} \times 4\Omega$$

When 5Ω is not there, 2Ω & 4Ω are in series with $12V$.

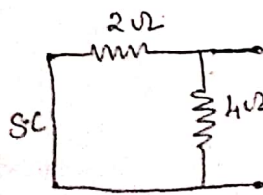
$$\therefore \text{Current flow through } 4\Omega = \frac{12}{2+4} = 2A$$

$$\text{Then } V_{TH} = V_{ab} = 2 \times 4 = \underline{\underline{8 \text{ Volts}}}$$

To find R_{TH}

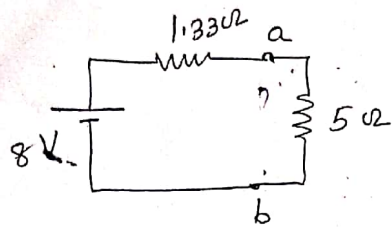
R_{TH} is the internal resistance of the circuit as viewed from terminals a & b with the voltage sources ~~replaced~~ short circuited

Then the circuit becomes



$$\Rightarrow R_{TH}, \quad R_{TH} = 2\Omega \parallel 4\Omega$$

$$= \frac{2 \times 4}{2+4} = \frac{8}{6} = \underline{\underline{1.33\Omega}}$$

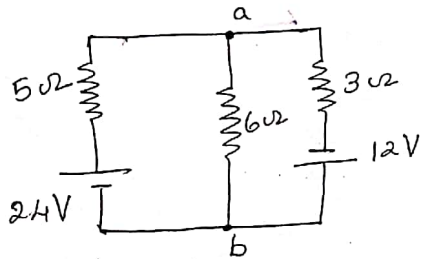


To determine the current through 5Ω , connect the 5Ω resistor back to terminal a & b.

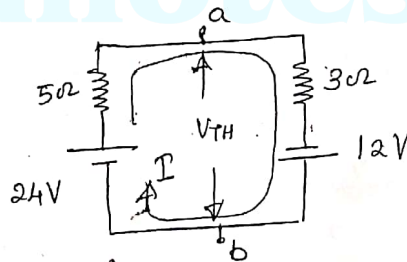
$$\text{The current through } 5\Omega = \frac{8}{5 + 1.33} = \underline{\underline{1.26A}}$$

Examples

- ① Determine the current through 6Ω resistor.



To find V_{TH} , disconnect 6Ω and redraw the circuit as shown below.



$$V_{TH} = V_{ab} = 24 - 5 \times I$$

$$= -12 + 3 \times I$$

$$\text{where } I = \frac{24 + 12}{5 + 3}$$

$$= \underline{\underline{4.5A}}$$

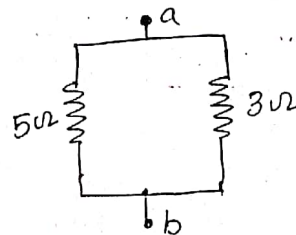
$$\therefore V_{TH} = 24 - 5 \times 4.5 = \underline{\underline{1.5V}}$$

To determine R_{TH} , replace $12V$ & $24V$ by short circuit

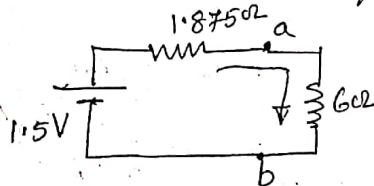
Then the circuit becomes

$$\text{Then, } R_{TH} = 5\Omega \parallel 3\Omega$$

$$= \frac{5 \times 3}{5 + 3} = \underline{\underline{1.875\Omega}}$$



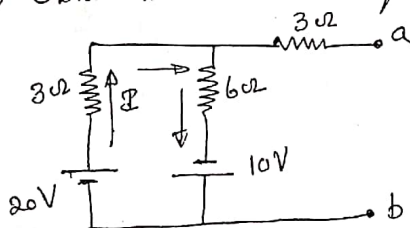
Then Thevenin's equivalent circuit can be redrawn as (15)



Then connect back the 6Ω resistance through which the current is to be determined.

$$\text{Then current through } 6\Omega = \frac{1.5}{6 + 1.875} = \underline{\underline{0.19 \text{ A}}}$$

② Obtain the Thevenin's equivalent circuit of fig-shown below.



Thevenin's theorem replaces this circuit by

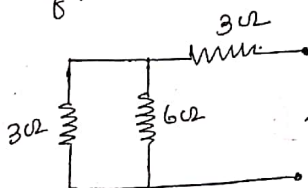
To find V_{TH}

Since no current passes through the upper 3Ω resistor,

$$V_{ab} = 20 - 3I \quad \text{or} \quad -10 + 6I, \quad \text{where } I = \frac{20 + 10}{3 + 6} = \underline{\underline{\frac{10}{3} \text{ A}}}$$

$$\therefore V_{ab} = V_{TH} = 20 - 3 \times \frac{10}{3} = \underline{\underline{10 \text{ V}}}$$

To find R_{TH} - short circuit voltage sources (Open circuit current sources)
of 20V & 10V. Then the circuit becomes

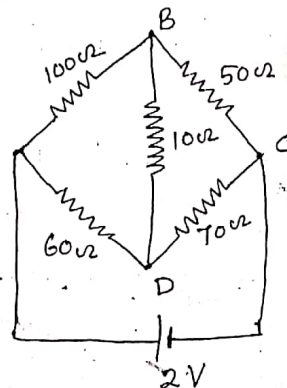


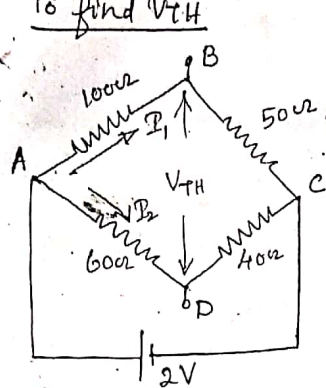
$$R_{TH} = 3 + 6 \parallel 3$$

$$= 3 + \frac{6 \times 3}{6 + 3} = \underline{\underline{5\Omega}}$$

③ Use Thevenin's theorem to determine the current through the 10Ω resistor in the circuit.

Now disconnect the 10Ω resistor
and obtain the Thevenin's equivalent circuit ^{viewed from} ~~at~~ B & D.





Here $V_{TH} = V_{BD}$

$= V_{AB} - V_{AD}$ (w.r.t. terminal A)

or $= V_{CB} - V_{CD}$ (w.r.t. terminal C)

Here paths ABC & ADC are in parallel. Let I_1 be the current through path ABC & I_2 be the current in path ADC.

Now, $V_{AB} = 100 \times I_1$ & $V_{AD} = 60 \times I_2$

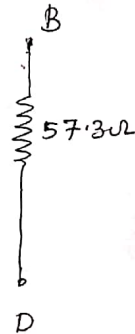
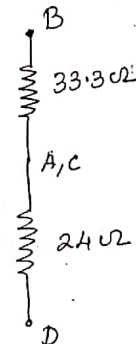
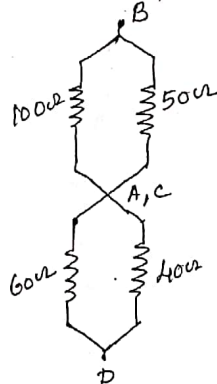
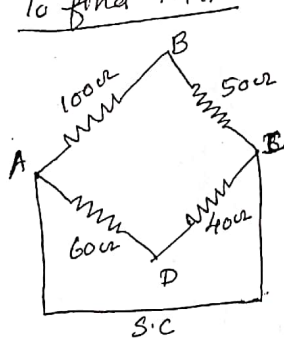
where $I_1 = \frac{2}{100+50} = 0.0133A$ & $I_2 = \frac{2}{60+40} = 0.02A$

Then $V_{BD} = V_{AB} - V_{AD}$

$= 100I_1 - 60I_2 = 1.33 - 1.2 = 0.13 \text{ Volts}$

ie $V_{TH} = V_{BD} = 0.13 \text{ Volts}$

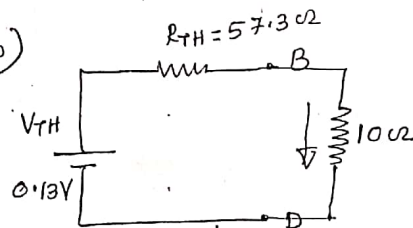
To find R_{TH} , short circuit the voltage source.



$R_{TH} = R_{BD} = (100 \parallel 50) + (60 \parallel 40)$

$= 33.3\Omega + 24\Omega$

$= 57.3\Omega$

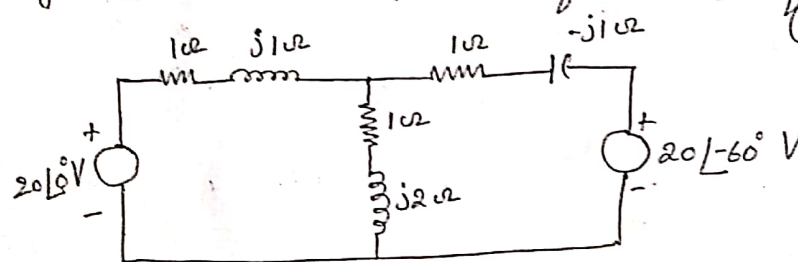


Then $I_{10\Omega} = \frac{V_{TH}}{R_{TH} + 10\Omega} = \frac{0.13}{57.3 + 10} = 1.93 \text{ mA}$

Theremin's Theorem (as applied to ac circuits)

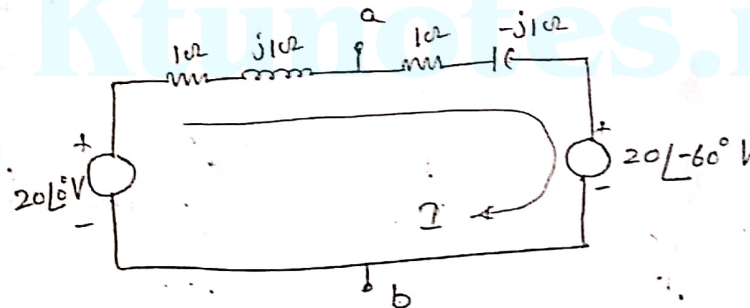
Any two terminal linear bilateral network containing impedances and energy sources may be replaced by a voltage source (V_{TH}) in series with an internal impedance (Z_{TH}). V_{TH} is the o.c. voltage at the terminals and Z_{TH} is the impedance viewed at the terminals when all the energy ^(voltage) sources are short-circuited (current sources open-circuited).

eg. Obtain Theremin's equivalent for the circuit shown below and find current through $(1+j2)\Omega$



To find V_{TH}

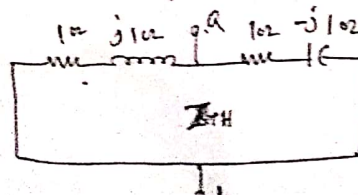
Disconnect $(1+j2)\Omega$ and the circuit then becomes

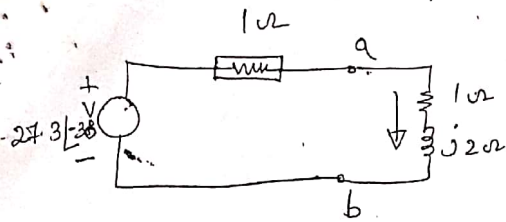


$$I = \frac{20\angle 0^\circ - 20\angle -60^\circ}{2} = 10\angle 60^\circ \text{ A}$$

$$V_{TH} = V_{ab} = 20\angle 0^\circ - 10\angle 60^\circ \times (1+j1) = \underline{\underline{27.3\angle -30^\circ \text{ V}}}$$

$$Z_{TH} = (1+j1) \parallel (1-j1) = \frac{(1+j1)(1-j1)}{2} = \frac{1+1}{2} = \underline{\underline{1\Omega}}$$

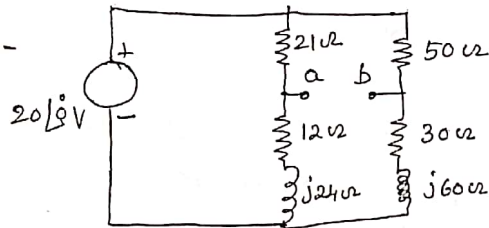




Then current through $(1+j2)\Omega = \frac{27.3 \angle -30^\circ}{1+1+j2}$

=

Exercise - Find Thevenin's equivalent circuit



Ans:

$$V_{TH} = 0.326 \angle 169.4^\circ \text{ V}$$

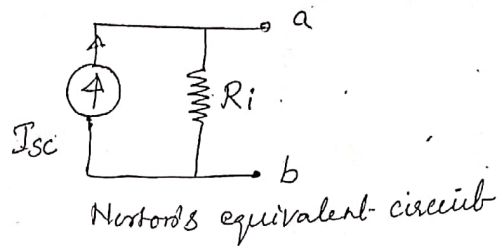
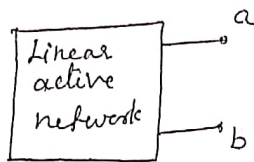
$$Z_{TH} = 47.35 \angle 6.8^\circ \Omega$$

Norton's theorem (DC circuits)

This theorem is an alternative to the Thevenin's theorem. Thevenin's theorem reduces a two-terminal active network to an equivalent constant voltage source and a series resistance whereas Norton's theorem replaces the network by an equivalent constant current source.

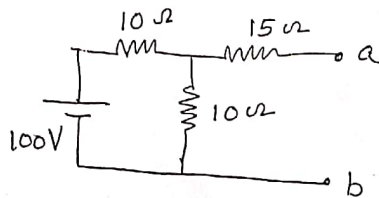
This theorem may be stated as follows:

Any two terminal active network containing voltage sources and resistances when viewed from its output terminals is equivalent to a constant ~~source~~ current source and a parallel resistance. The constant current is equal to the current which flows in a short circuit placed between the terminals and parallel resistance is equal to the resistance of the network when open circuited.



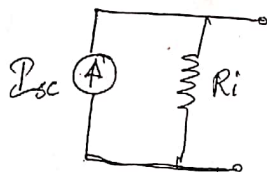
Explanation:

Consider the circuit shown below.

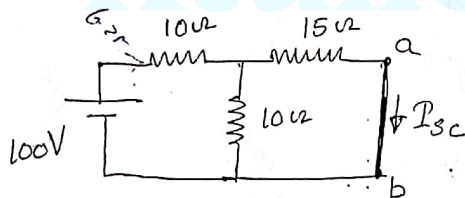


Obtain Norton's equivalent circuit

Norton's theorem replaces this circuit by



To find I_{sc} , short circuit terminals a & b as shown below.



When a & b are short circuit
total circuit resistance
consists of 10Ω in series

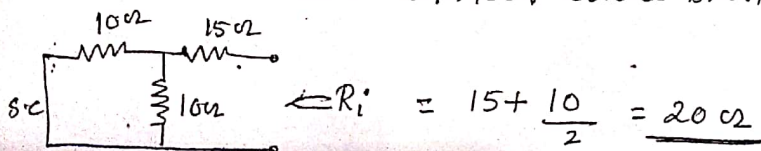
with the parallel combination of 10Ω & 15Ω.

$$\therefore \text{Total circuit resistance, } R_T = 10\Omega + \frac{15 \times 10}{15 + 10} = 16\Omega$$

$$\text{Then, total supply current, } I_T = \frac{100}{16} = 6.25\text{ A}$$

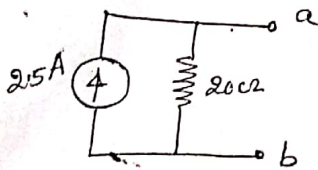
$$\text{Then } I_{sc} = I_{15\Omega} = I_T \times \frac{10}{10+15} = 6.25 \times \frac{10}{25} = 2.5\text{ A}$$

To find R_i (It is the internal resistance as viewed from a & b with 100V source short circuited)

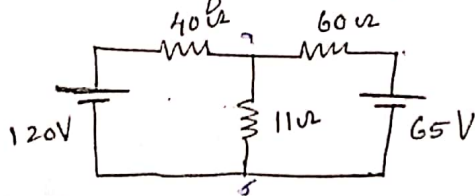


$$\leftarrow R_i = 15 + \frac{10}{2} = 20\Omega$$

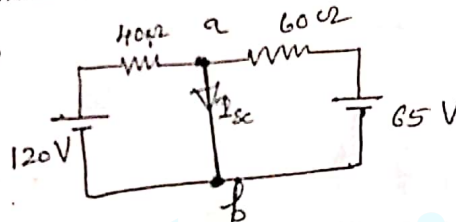
Then the Norton's equivalent circuit can be drawn as



Example 2 . By means of Norton's theorem, find the current flow through 11Ω resistance

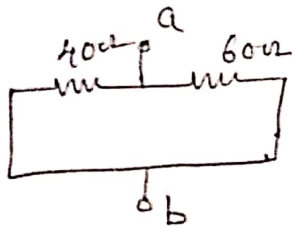


To find I_{sc} , Disconnect 11Ω and short circuit the terminals. Then the circuit becomes



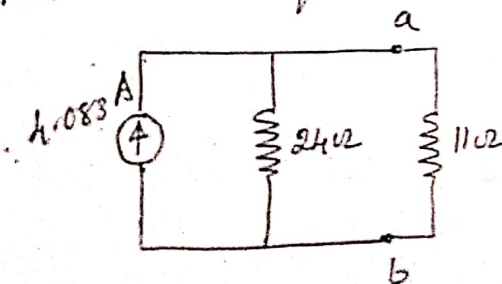
$$\text{Now } I_{sc} = \frac{120}{40} + \frac{65}{60} = 3 + 1.083 = \underline{\underline{4.083 \text{ A}}}$$

To find R_i , Remove all battery sources, then the circuit becomes



$$\begin{aligned} \therefore R_i &= (40 \parallel 60)\Omega \\ &= \frac{40 \times 60}{40 + 60} = \underline{\underline{24\Omega}} \end{aligned}$$

Then Norton's equivalent circuit is

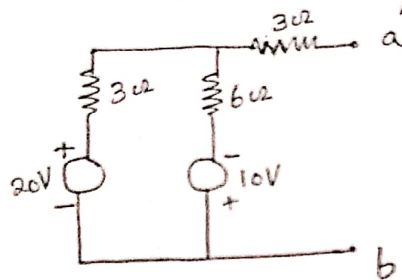


Connect back 11Ω resistance across a & b

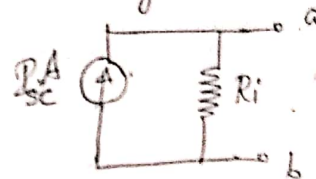
$$\begin{aligned} \text{The } I_{11\Omega} &= \frac{4.083 \times 24}{24 + 11} \\ &= \underline{\underline{2.8 \text{ A}}} \end{aligned}$$

Ex 3 Obtain Norton's equivalent circuit

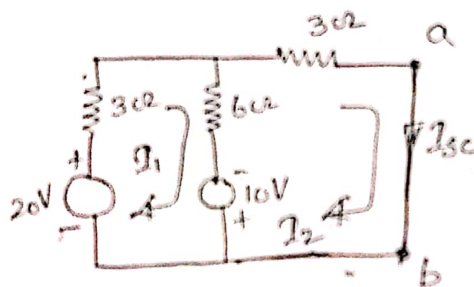
(21)



Norton's Theorem replaces this network by



To find I_{sc} , short circuit a & b . Then the circuit becomes

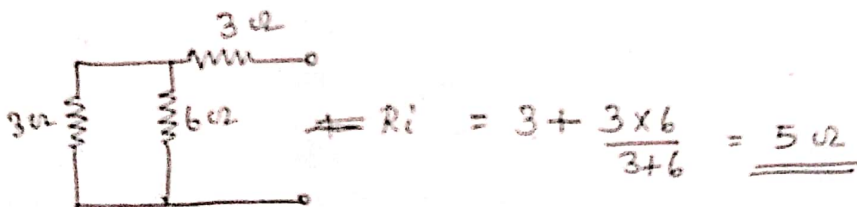


then I_{sc} can be found out by mesh method. The matrix equation is

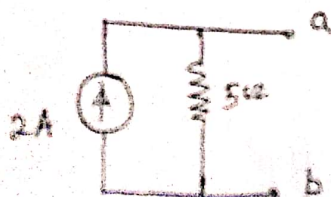
$$\begin{bmatrix} 9 & -6 \\ -6 & 9 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 30 \\ -10 \end{bmatrix}$$

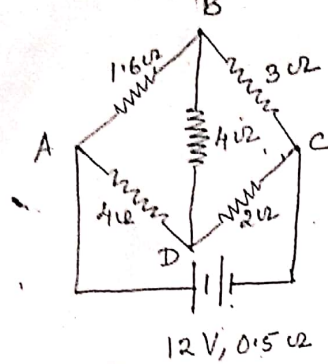
On solving, $I_{sc} = I_2 = \frac{\begin{vmatrix} 9 & 30 \\ -6 & -10 \end{vmatrix}}{\begin{vmatrix} 9 & -6 \\ -6 & 9 \end{vmatrix}} = \frac{-90 + 180}{81 - 36} = \underline{\underline{2 \text{ A}}}$

To find R_i , short circuit all voltage sources, then the circuit is redrawn as



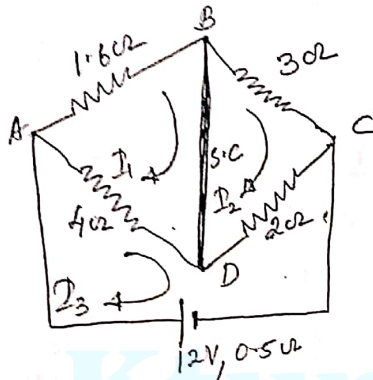
Then the Norton's equivalent circuit can be drawn as





Obtain the current flow through 4Ω (I_{BD}) by applying Norton's ~~equivalent~~ theorem

To find I_{sc} - Disconnect 4Ω resistor connected across BD and short-circuit BD



Assign loop currents I_1 & I_2 & I_3

$$\text{Then } I_{sc} = I_1 - I_2$$

The matrix equation can be written as

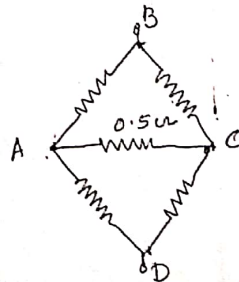
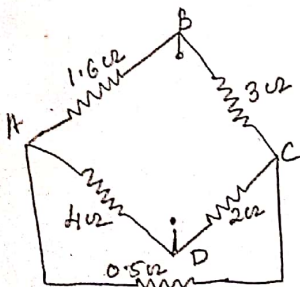
$$\begin{bmatrix} 5.6 & 0 & -4 \\ 0 & 5 & -2 \\ -4 & -2 & 6.5 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 12 \end{bmatrix}$$

on solving $I_1 = \frac{\Delta_1}{\Delta} = \frac{-240}{-79.6} = \underline{\underline{3.01 A}}$

$$I_2 = \frac{\Delta_2}{\Delta} = \frac{-134.4}{-79.6} = \underline{\underline{1.688 A}}$$

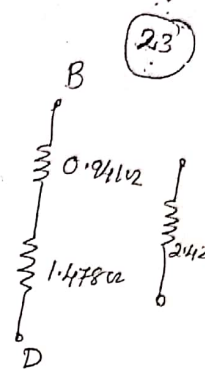
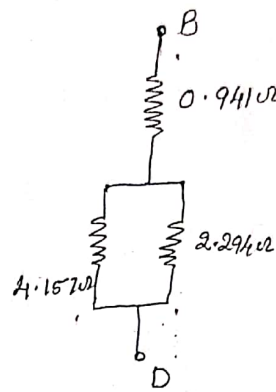
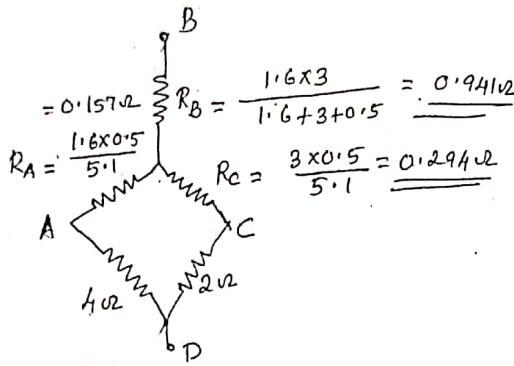
$$I_{sc} = (I_1 - I_2) = \underline{\underline{1.32 A}}$$

To find R_i , s.c, 12V battery leaving its internal resistance

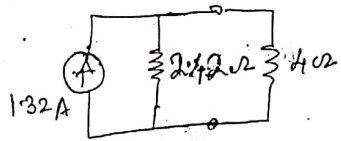


To obtain R_i ,
i.e. the resistance between B & D,
convert one of the delta into star,
say ΔABC to star

Then the circuit becomes



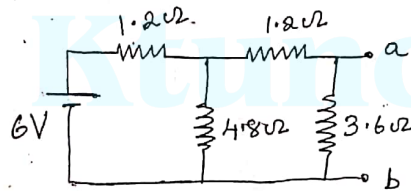
Norton's equivalent circuit



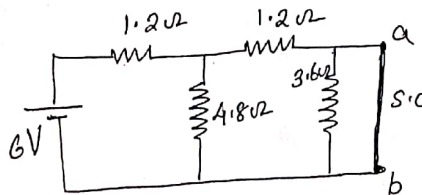
$$I_{A2} = 1.32 \times \frac{2.42}{6.42} = 0.5 A$$

eg.5

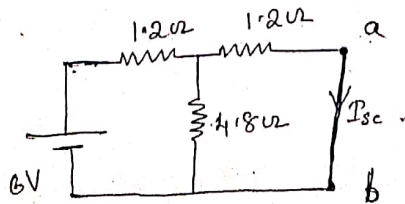
Find ~~thevenin~~ Norton's equivalent circuit & hence obtain thevenin's equivalent circuit



To find I_{sc} , short circuit a & b.



when a b is short-circuited, no current flows through 3.6 ohm $\therefore$ the circuit is reduced to



$$R_T = 1.2 + (4.8 \parallel 1.2) \Omega$$

$$= 1.2 + \frac{4.8 \times 1.2}{6} = 2.16 \Omega$$

$$\therefore I_T = \frac{V}{R_T} = \frac{6}{2.16} = 2.77 A$$

$$\therefore I_{sc} = I_T \times \frac{4.8}{4.8 + 1.2} = 2.77 \times \frac{4.8}{6} = 2.22 A$$

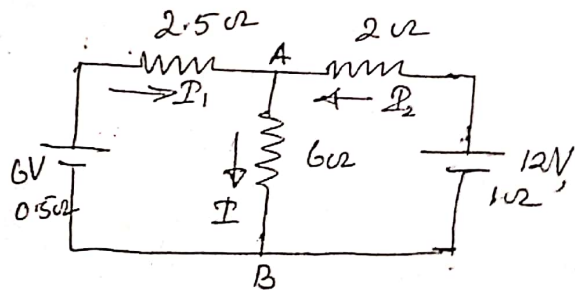
Superposition theorem

In a linear network of resistances containing more than one source of emf, the current which flows in any branch is the ^{algebraic} sum of all the currents which ^{would} flow in that branch if each voltage source is considered separately and all other voltage sources replaced ~~by~~ for the time being by short circuits. (Current sources by open circuits)

$\therefore$ According to this theorem, if there are a number of emfs acting simultaneously in any linear network, then each emf acts independently of the others as if other emfs did not exist. The value of current in any conductor is the algebraic sum of the current due to each emf while acting alone.

i.e. current in or voltage across any conductor is obtained by superposing the currents and voltages due to each emf in the network.

Consider the network shown here.
Find the current through AB by applying principle of superposition.

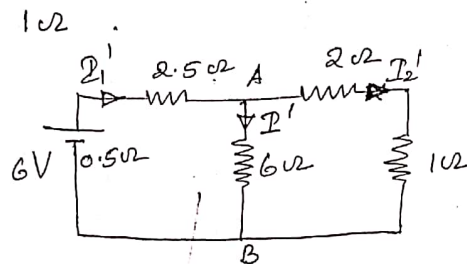


Here I_1 , I_2 & I represent the values of currents which are

due to simultaneous action of the two sources. (25)

Case ① - when 6V battery is acting alone

In this case, remove 12V battery leaving the internal resistance



Let I_1' , I_2' & I' be the currents in the branches

There are two parallel paths between A & B. $\therefore$

$$\therefore \text{Total resistance } R_T = 2.5 + 0.5 + \frac{6 \times 3}{6+3}$$

$$= \underline{\underline{5 \Omega}}$$

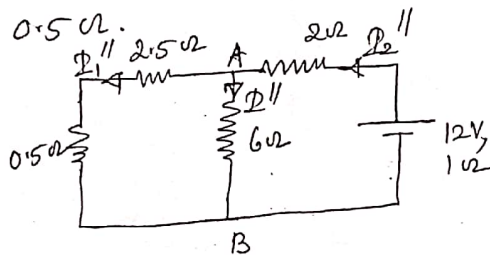
$$\text{Then total current, } I_1' = \frac{V}{R_T} = \frac{6}{5} = \underline{\underline{1.2 A}}$$

This current divides at A into I' & I_2'

$$\therefore I' = 1.2 \times \frac{3}{6+3} = \underline{\underline{0.4 A}} \quad \& \quad I_2' = 1.2 - 0.4 = \underline{\underline{0.8 A}}$$

Case ② when 12V battery acts alone.

Now remove 6V battery, leaving its internal resistance



Let I_1'' , I_2'' & I'' be the currents in this case

$$\text{Now } R_T = 1 + 2 + \frac{6 \times 3}{6+3} = \underline{\underline{5 \Omega}}$$

$$\text{Then } I_2'' = \frac{12}{5} = \underline{\underline{2.4 A}}$$

At point A, current divides into I_1'' & I''

$$\therefore I_1'' = 2.4 \times \frac{6}{9} = \underline{\underline{1.6 A}} \quad \& \quad I'' = 2.4 - 1.6 = \underline{\underline{0.8 A}}$$

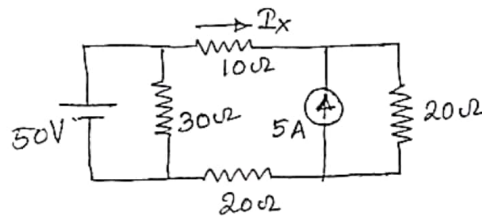
found by algebraically adding the currents due to each emf.

$$\therefore I_1 = I_1' - I_1'' =$$

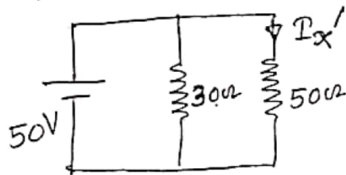
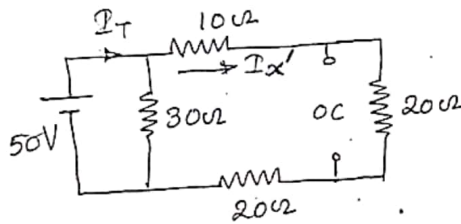
$$I_2 = I_2'' - I_2' =$$

$$I_{5\Omega} = I = I' + I'' = \underline{\underline{1.2A}}$$

Ex.1 Obtain the current I_x in the 10Ω resistance using superposition theorem



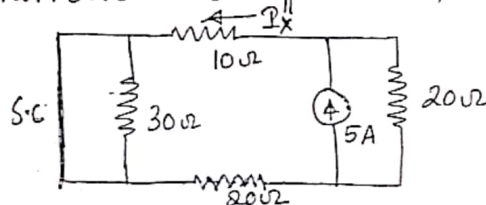
when 50V source acts alone - Open circuit the 5A current source. Then the circuit becomes



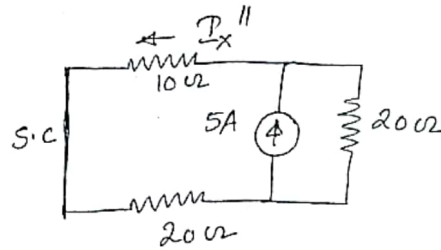
$$I_x' = \frac{50}{50} = \underline{\underline{1A}}$$

when 5A current source acts alone

Short-circuit 50V source, then the circuit becomes



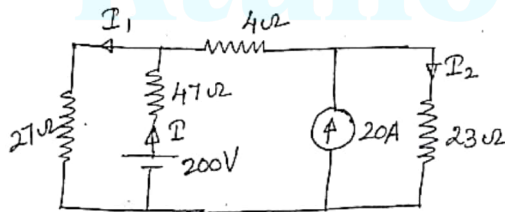
when 50V source is short circuited, no current flows through 30Ω & hence can be removed. (2.7)



$$\text{Then } I_x'' = \frac{5 \times 20}{20 + 30} = 2A$$

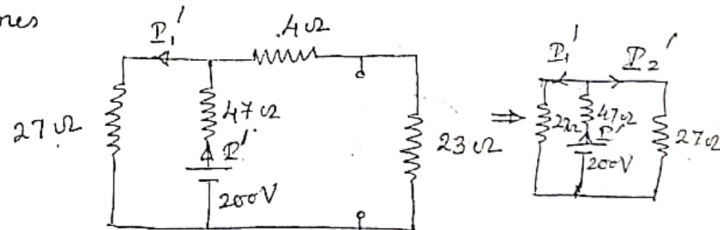
$$\begin{aligned} \text{Then the actual current flow through } 10\Omega &= I_x' - I_x'' \\ &= \underline{\underline{-1A}} \end{aligned}$$

Q2 Compute the current in the 23Ω resistor by applying the superposition principle.

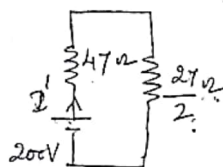


when 200V acts alone, ~~replace~~ open circuit 20A current source.

Then the circuit becomes



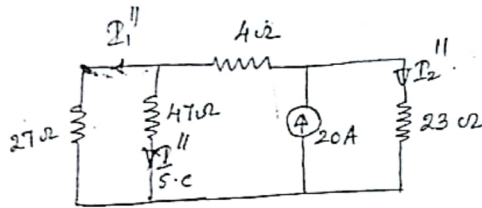
Now 27Ω and 47Ω are in parallel



$$\text{Now } I_1' = \frac{200}{47 + 13.5} = \underline{\underline{3.31A}}$$

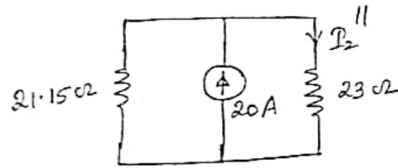
$$\text{Then } I_2' = \frac{3.31A}{2} = \underline{\underline{1.65A}}$$

When 20 A current source acts alone, short-circuit the 200 V source. Then the circuit becomes



To the left of 20 A source

$$R_T = 4 + \frac{27 \times 47}{27 + 47} = 21.15 \Omega$$



$$I_2'' = \frac{20 \times 21.15}{23 + 21.15} = 9.58 \text{ A}$$

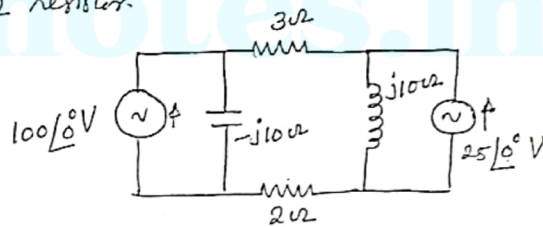
∴ Actual current in the 23Ω resistor, $I_2 = I_2' + I_2''$

$$= 1.65 + 9.58$$

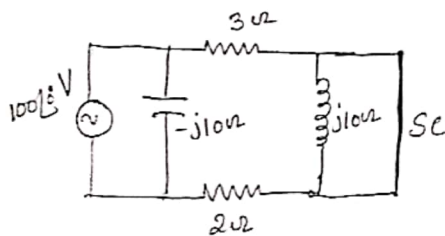
$$= 11.23 \text{ A}$$

Ex. 3

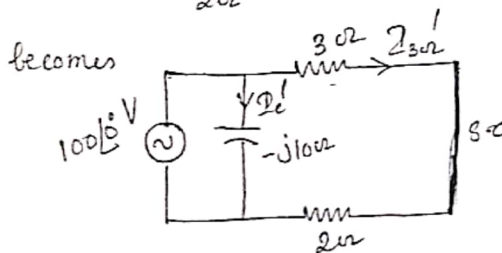
By superposition principle, find the current in the capacitor of the circuit and 3Ω resistor.



When 100∠0° V acts alone



Since no current flows through j10Ω (∵ a dead short circuit across j10Ω) j10Ω has no effect and hence can be removed. ∴ The circuit



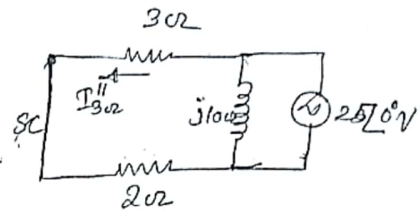
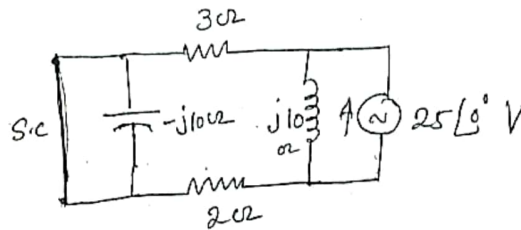
$$\text{then } I_{3\Omega}' = \frac{100 \angle 0^\circ}{2 + 3} = 20 \angle 0^\circ \text{ A}$$

$$\therefore I_C' = \frac{100 \angle 0^\circ}{-j10} = \frac{100 \angle 0^\circ}{10 \angle -90^\circ}$$

$$= 10 \angle 90^\circ \text{ A} = j10 \text{ A}$$

when $25\angle 0^\circ$ Volts acts alone

(29)



$$I_c'' = 0 \text{ A} \quad \& \quad I_{3\Omega}'' = \frac{25\angle 0^\circ}{3+2} = 5\angle 0^\circ \text{ A}$$

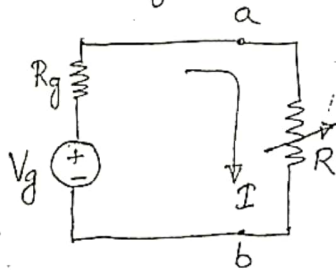
$$\therefore I_c = I_c' + I_c'' = 10\angle 90^\circ \text{ A} \quad \& \quad P_{3\Omega} = P_{3\Omega}' - P_{3\Omega}''$$

$$= 20\angle 0^\circ - 5\angle 0^\circ$$

$$= 15\angle 0^\circ \text{ A}$$

Maximum Power transfer theorem (as applied to DC circuits)

Maximum power will be delivered from a network to a load, when load resistance is equal to the internal resistance of the network as viewed from the output terminals with all sources of emf short-circuited (Current sources open-circuited)



'R' is the load resistance, connected across the terminals a & b of the network which consists of a voltage source 'V_g' and an internal resistance R_g.

According to this theorem, R will absorb maximum power from the network when $R = R_g$.

Proof

Here load current, $I = \frac{V_g}{R_g + R}$ & power delivered to

the load $P = I^2 R$

$$= \left(\frac{V_g}{R_g + R} \right)^2 \times R$$

Power delivered is maximum, when $\frac{dP}{dR} = 0$

(30)

$$\frac{dP}{dR} = \frac{d}{dR} \left[V_g^2 \cdot \frac{R}{(R_g + R)^2} \right] = 0$$

$$\frac{dP}{dR} = V_g^2 \left[\frac{(R_g + R)^2 \cdot 1 - R \cdot 2(R_g + R)}{(R_g + R)^4} \right] = 0$$

$$\text{or } (R_g + R)^2 - 2R(R_g + R) = 0$$

$$\text{or } (R_g + R)^2 = 2R(R_g + R)$$

$$R_g + R = 2R$$

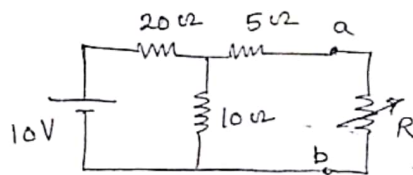
$$\text{Then, } \underline{\underline{R = R_g}}$$

i.e. Power delivered is maximum, when $R = R_g$

$$\text{and maximum power, } P_{\max} = \left(\frac{V_g}{R_g + R} \right)^2 \times R = \left(\frac{V_g}{2R} \right)^2 \times R$$

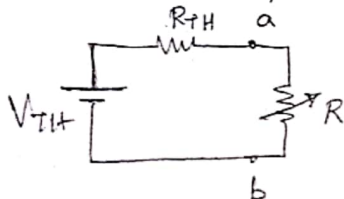
$$= \frac{V_g^2}{4R} \text{ watts}$$

eg. In the circuit shown, Calculate the resistance 'R' which will allow maximum power dissipation. Also calculate the maximum power.

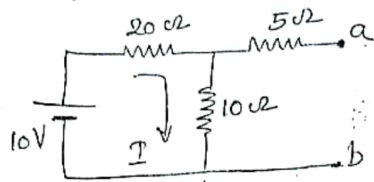


'R' will abstract maximum power, when $R = R_i$

∴ To find R_i , the network to the left of the terminals 'ab' can be replaced by a Thevenin's equivalent circuit.



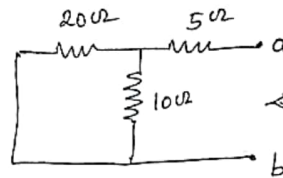
To find V_{TH} , disconnect R and the circuit becomes



$V_{TH} = V_{ab} = V_{10\Omega} = 10 \times I$ Volts, where

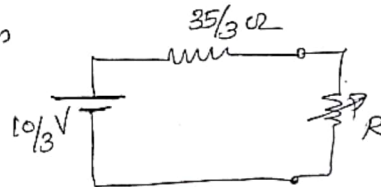
$$I = \frac{10}{20+10} = \frac{1}{3} \text{ A}, \text{ then } V_{TH} = 10 \times \frac{1}{3} = \underline{\underline{\frac{10}{3} \text{ Volts}}}$$

To find R_{TH}



$$R_{TH} = 5 + \frac{20 \times 10}{20+10} = \underline{\underline{\frac{35}{3} \Omega}}$$

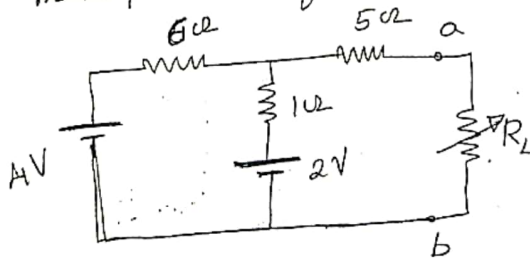
Now the circuit becomes



For the maximum power dissipation, $R = R_i = \underline{\underline{\frac{35}{3} \Omega}}$

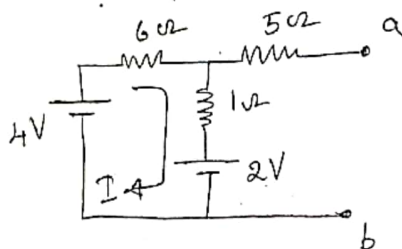
$$\therefore \text{Max power} = \frac{V^2}{4R} = \frac{\left(\frac{10}{3}\right)^2}{4 \times \frac{35}{3}} = \underline{\underline{\frac{5}{21} \text{ watts}}}$$

Ex2. For the circuit shown, calculate the load resistance for max power transfer and the value of this power.



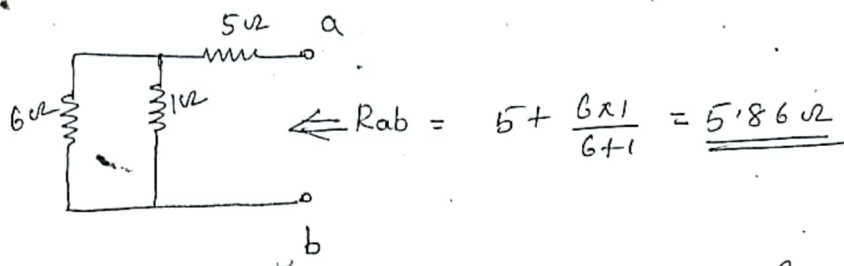
Using Thevenin's theorem reduce the circuit to a single voltage source and resistance in series with it.

$\therefore$ Disconnect load resistance. Then the circuit becomes

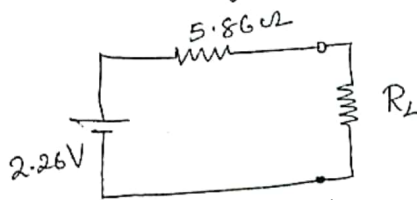


$$I = \frac{4-2}{4+2+1} = \frac{2}{7} = \underline{\underline{0.29 \text{ A}}}$$

$$V_{ab} = 4 - 6 \times 0.29 = \underline{\underline{2.26 \text{ V}}}$$



Then the original circuit can be replaced by

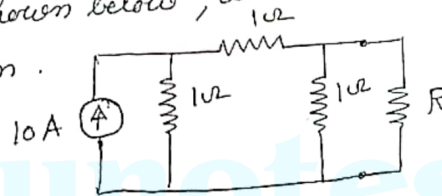


when $R_L = R_i = 5.86 \Omega$,
Power becomes maximum.

$$\therefore P_{max} = \frac{(2.26)^2}{4 \times 5.86} = \underline{\underline{0.22 \text{ Watts}}}$$

Exercise

In the circuit shown below, what is the value of R when power is maximum.



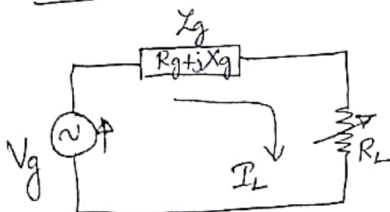
(Ans: $R = \frac{2}{3} \Omega$
 $P = \frac{25}{6} \text{ Watts}$)

Maximum power transfer theorem (as applied to ac circuits)

In DC circuits, the load is a pure resistance and this load will absorb max-power when $R_L = R_i$.

But in ac circuits, the load may be a pure resistance or a complex impedance. So we shall consider the two cases separately.

Case ① - Variable pure resistive load



when $\frac{dP_L}{dR_L} = 0$

Here load current, $I_L = \frac{V_g}{\sqrt{(R_g + R_L)^2 + X_g^2}}$

Power dissipated in the load,

$P_L = I_L^2 R_L$ and P_L is maximum

$$= \frac{V_g^2}{(R_g + R_L)^2 + X_g^2} \cdot R_L$$

$$\text{ie } \frac{V_g^2 \cdot [(R_g + R_L)^2 + X_g^2 - R_L \cdot 2(R_g + R_L)]}{[(R_g + R_L)^2 + X_g^2]^2} = 0$$

$$\text{or } (R_g + R_L)^2 + X_g^2 - 2R_L R_g - 2R_L^2 = 0$$

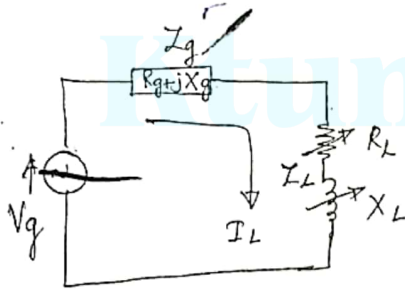
$$R_g^2 + R_L^2 + 2R_g R_L + X_g^2 - 2R_L R_g - 2R_L^2 = 0$$

$$R_g^2 + X_g^2 = R_L^2 \quad \text{or}$$

$$R_L = \sqrt{R_g^2 + X_g^2} = |Z_g|$$

Thus for maximum power transfer to the load, the load resistance must be equal to the magnitude of the source impedance. If $X_g = 0$, $R_L = R_g$ as in the DC case.

Case ② - Load is a complex impedance



$$\text{Here } I_L = \frac{V_g}{\sqrt{(R_g + R_L)^2 + (X_g + X_L)^2}}$$

$$P_L = I_L^2 R_L = \frac{V_g^2}{(R_g + R_L)^2 + (X_g + X_L)^2} \cdot R_L$$

Here P_L is maximum when the partial derivatives

$$\frac{\partial P_L}{\partial R_L} = 0 \quad \text{and} \quad \frac{\partial P_L}{\partial X_L} = 0$$

$$\frac{\partial P_L}{\partial X_L} = 0 \quad \text{gives } X_L = -X_g \quad \text{Substituting this value of } X_L$$

$$\text{in } \frac{\partial P_L}{\partial R_L} = 0 \quad \text{gives } \underline{R_L = R_g} \quad \text{Hence for max power}$$

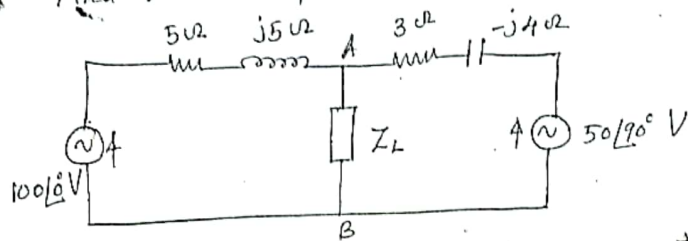
$$\text{transfer condition, } R_L + jX_L = R_g - jX_g$$

$$\text{ie } Z_L = Z_g^* \quad \text{ie the load}$$

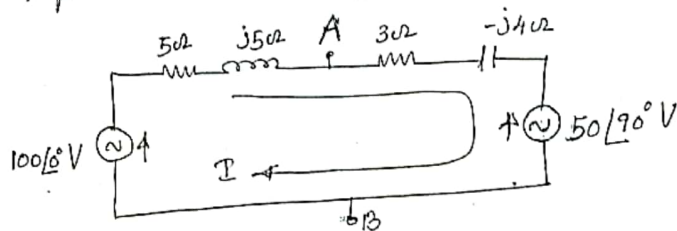
impedance must equal to the conjugate of the internal impedance

if X_L is fixed, power transfer, $R_L = \sqrt{R_g^2 + (X_g + X_L)^2}$.

Ex. 1 Find the max. power transferred to the load Z_L .



Power delivered is maximum, when $Z_L = Z_i^*$. To find Z_i , replace the network by Thevenin's equivalent circuit about AB.

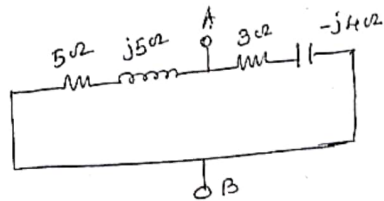


$$I = \frac{100\angle 0^\circ - 50\angle 90^\circ}{8 + j1}$$

$$V_{AB} = 100\angle 0^\circ - \left(\frac{100 - j5}{8 + j1} \right) (5 + j5) = \frac{50 - j250}{8 + j1}$$

$$= \underline{\underline{19.615 \angle -78.7^\circ \text{ V}}}$$

To find Z_i



$$Z_{TH} = (5 + j5) \parallel (3 - j4)$$

$$= \frac{(5 + j5)(3 - j4)}{8 + j1} = \frac{55 - j15}{13}$$

$$= \underline{\underline{4.384 \angle -15.25^\circ \Omega}}$$

Power reaches max. value when $Z_L = Z_i^*$
ie when $Z_L = \frac{55 + j15}{13} \Omega$

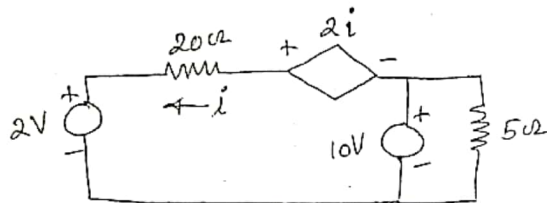
$$I_L = \frac{V_{TH}}{Z_{TH} + Z_L} = \frac{50 - j250}{13 \left(\frac{55 - j15}{13} + \frac{55 + j15}{13} \right)} = \frac{50 - j250}{110}$$

$$= 2.32 \angle -78.7^\circ \text{ A}$$

$$\text{Then } P_L = I^2 R_L = (2.32)^2 \times \frac{55}{13} = 22.73 \text{ Watts}$$

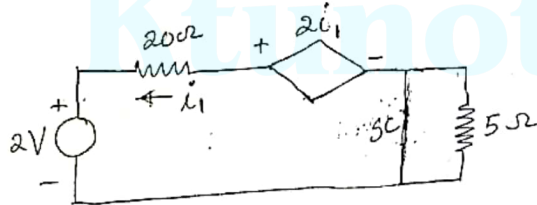
With dependent sources - Application of theorems

Ex. 4 By applying superposition theorem, determine the current i in the circuit shown below.



has 2
This circuit has independent sources and one dependent source.

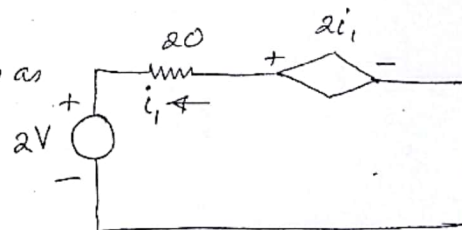
when 2V acts alone, short circuit 10V, then the circuit becomes



Since it is a dead short circuit across 5Ω, 5Ω can be removed.

Then the circuit can be redrawn as

Applying KVL around the loop,

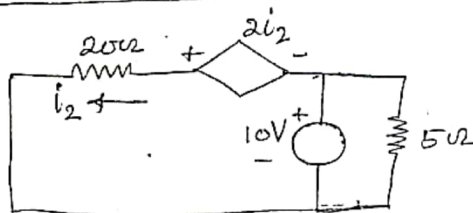


$$2 + 20i_1 - 2i_1 = 0$$

$$18i_1 = -2$$

$$i_1 = \frac{-2}{18} = -\frac{1}{9} \text{ A}$$

when 10V acts alone, short circuit 2V, then the circuit is



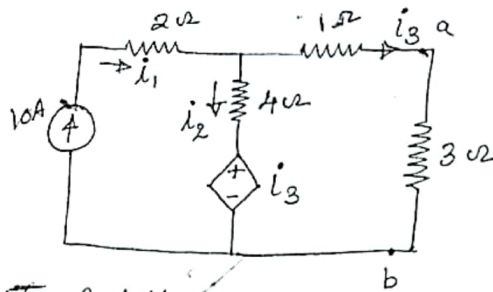
$$20i_2 - 2i_2 - 10 = 0$$

$$18i_2 = 10$$

$$i_2 = \frac{10}{18} = \frac{5}{9} \text{ A}$$

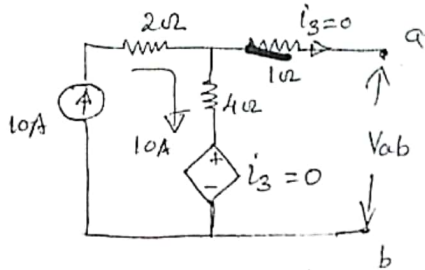
$$\therefore i = i_1 + i_2 = \frac{5}{9} - \frac{1}{9} = \frac{4}{9} \text{ A}$$

Fig. 2 Use Thevenin's theorem to find V_{TH} of the 3Ω resistor in the circuit shown below.



To find V_{TH}

To start with disconnect 3Ω resistor. Then the circuit becomes

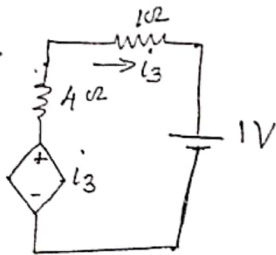


Then $i_3 = 0 \therefore$ Dependent source has zero voltage across it and hence voltage across ab is just the voltage across 4Ω resistor

$$\therefore V_{TH} = V_{ab} = V_{4\Omega} = 4 \times 10 = \underline{\underline{40V}}$$

To find R_{TH} : The best procedure with dependent source is to apply a voltage source to terminals and measure the current through this source. We must first remove all independent sources. (Open circuit current sources and short circuit voltage sources)

Then the circuit becomes



Applying KVL around the loop.

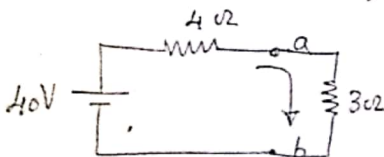
$$i_3 - 4i_3 - i_3 - 1 = 0$$

$$-4i_3 = 1 \quad \& \quad \underline{\underline{i_3 = -1/4 \text{ A}}}$$

$$\therefore R_{TH} = \frac{1}{1/4} = \underline{\underline{4\Omega}}$$

Then the Thevenin's equivalent can be drawn as

connect the 3Ω across $a \& b$



$$I_{3\Omega} = \frac{40}{4+3} = \frac{40}{7} \text{ A}$$

$$V_{3\Omega} = \frac{40}{7} \times 3 = \underline{\underline{\frac{360}{7} \text{ Volts}}}$$

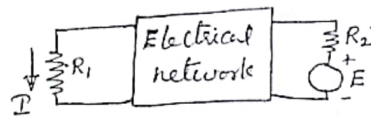
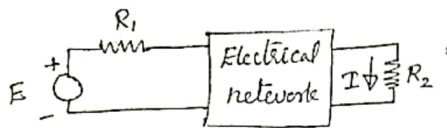
Reciprocity Theorem

In a linear active network, if a source of emf 'E' in any branch produces a current 'I' in any other branch, then the same emf 'E' acting in the second branch would produce the same current 'I' in the first branch. In other words, it simply means that $E \leftrightarrow I$ are mutually transferenceable.

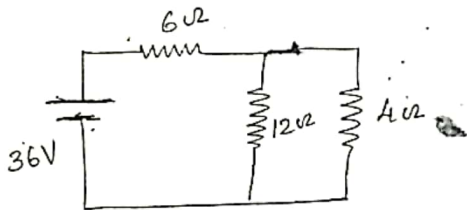
The ratio E/I is known as transfer resistance.

(or)

This theorem states that in a linear active network, a voltage source 'V' volts in a branch gives rise to a current 'I' in another branch, the ratio $\frac{V}{I}$ is constant, when the position of V & I are interchanged.



eg. In the network shown below, find the current in 4Ω resistance, and verify reciprocity theorem. Also calculate the transfer resistance.



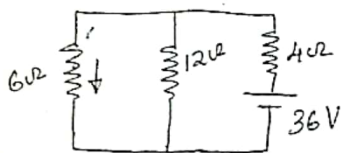
To find current through 4Ω

$$R_T = 6 + \frac{12 \times 4}{12 + 4} = 9\Omega$$

$$I_T = \frac{36}{9} = 4\text{ A}$$

$$I_{4\Omega} = 4 \times \frac{12}{16} = 3\text{ A}$$

When 36V battery is shifted to 4Ω branch,



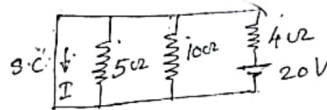
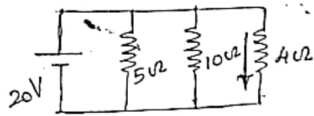
$$R_T = 4 + \frac{12 \times 6}{12 + 6} = 8\Omega$$

$$I_T = \frac{36}{8} = 4.5\text{ A}$$

$$I_{6\Omega} = 4.5 \times \frac{12}{18} = 3\text{ A}$$

$$\text{Transfer resistance} = \frac{36}{3} = 12\Omega$$

2. In the network shown, find the current through the resistor. Verify reciprocity theorem. What changes results in the current through the 5Ω and 10Ω branches.



$$I_{4\Omega} = \frac{20}{4} = 5A$$

$$I_{5\Omega} = \frac{20}{5} = 4A$$

$$I_{10\Omega} = \frac{20}{10} = 2A$$

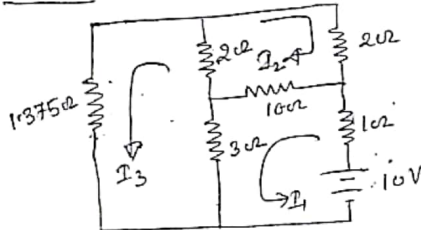
Since $I_{4\Omega} = I_{sc}$, theorem is verified.

$$I = \frac{20}{4} = 5A$$

$$I_{5\Omega} = 0$$

$$I_{10\Omega} = 0$$

Exercise

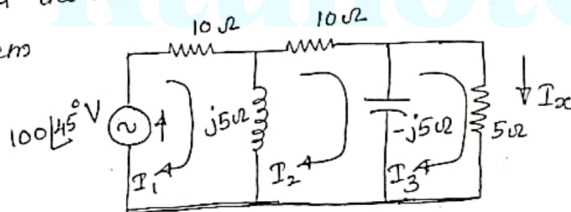


In this network, find I_3 and verify reciprocity theorem.

$$\begin{bmatrix} 14 & 10 & -3 \\ 10 & 14 & 2 \\ -3 & 2 & 6.375 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} 10 \\ 0 \\ 0 \end{bmatrix}$$

$$I_3 = \frac{\begin{vmatrix} 14 & 10 & 10 \\ 10 & 14 & 0 \\ -3 & 2 & 0 \end{vmatrix}}{\begin{vmatrix} 14 & 10 & -3 \\ 10 & 14 & 2 \\ -3 & 2 & 6.375 \end{vmatrix}} = 2A$$

Fig.3 Find the current I_x in the circuit and verify reciprocity theorem. To prove reciprocity



Find $I_x = I_3$

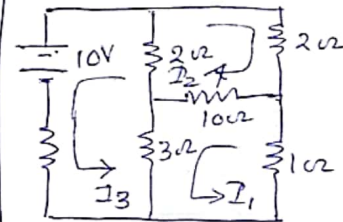
$$\begin{bmatrix} 10+j5 & -j5 & 0 \\ -j5 & 10 & j5 \\ 0 & j5 & 5-j5 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} 100\angle 45^\circ \\ 0 \\ 0 \end{bmatrix}$$

On solving, $I_x = I_3 = 2.69 \angle 69.5^\circ A$

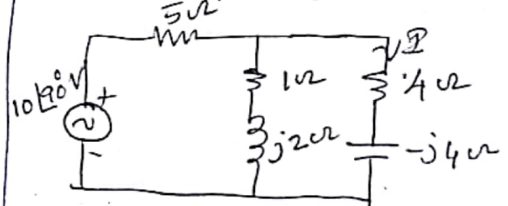
Limitations of Reciprocity theorem

1. It is applicable only to single source networks and not to a multisource network.
2. The network should not have any time varying elements.

Shift 10V to 1.375Ω branch

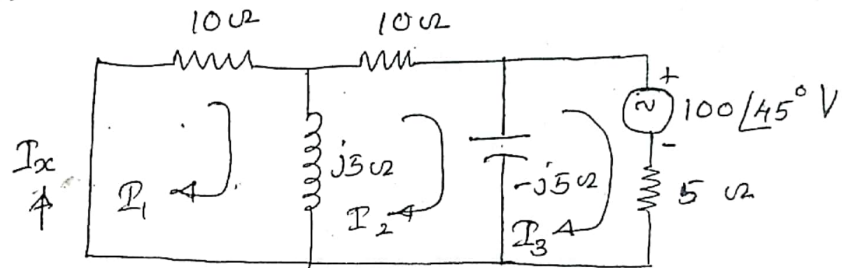


Verify reciprocity theorem



Ans 7 $I = (0.57 + j0.18) A$

when $100\angle 45^\circ V$ source is shifted to 5Ω branch, the circuit becomes



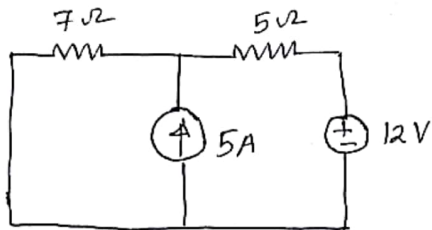
Here find $I_x = I_1$,

$$\begin{bmatrix} 10+j5 & -j5 & 0 \\ -j5 & 10 & j5 \\ 0 & j5 & 5-j5 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 100\angle 45^\circ \end{bmatrix}$$

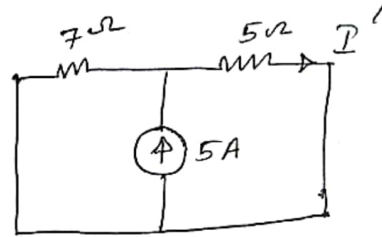
$$I_x = I_1 = \underline{\underline{2.69\angle 69.5^\circ A}}$$

The value of I_x is the same in these two cases. Thus the theorem is verified.

- 1) Find the current flowing through the 5Ω resistor shown in fig. if all active elements are ideal. Use superposition theorem.

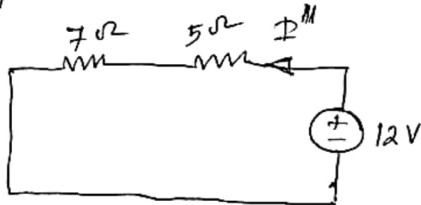


Soln: when 5A acts alone



$$I' = \frac{5 \times 7}{12} = 2.91 \text{ A}$$

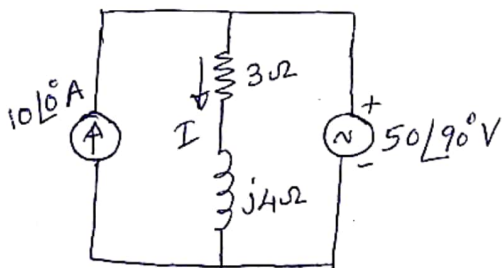
when 12V acts alone
open circuit current source



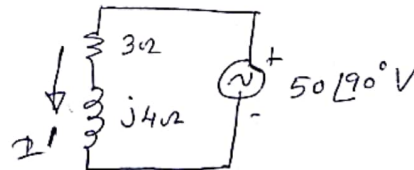
$$I'' = \frac{12}{12} = 1 \text{ A}$$

when both the sources acts, $I = I' - I'' = 1.91 \text{ A}$

- 2) Use superposition theorem to find the current I in the circuit



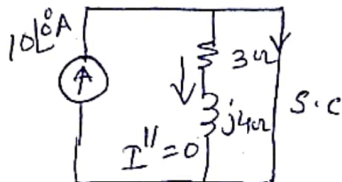
when ~~50∠90° V~~ acts alone
O.C current source



$$I' = \frac{50 \angle 90^\circ}{3 + j4} = \frac{50 \angle 90^\circ}{5 \angle 53.13^\circ}$$

$$= 10 \angle 36.87^\circ \text{ A}$$

when $10 \angle 0^\circ \text{ A}$ acts
S.C. voltage source.



No current flows through $(3 + j4)\Omega$, as there is a ^{lead} short circuit across $(3 + j4)\Omega$. Hence $I'' = 0$

Then with both the sources, $I = I' + I'' = 10 \angle 36.87^\circ \text{ A}$

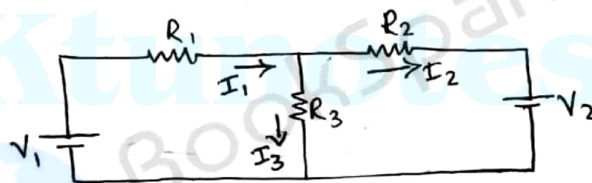
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Module 1

Network theorems.

1) Superposition theorem:

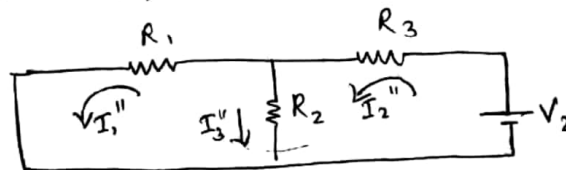
It states that, in a linear bi-lateral network consisting of more than one source, the effect on each parameter in the network with all the sources will be equal to the algebraic sum of the effects taking each source separately.



Consider V_1 ,



Consider V_2 ,

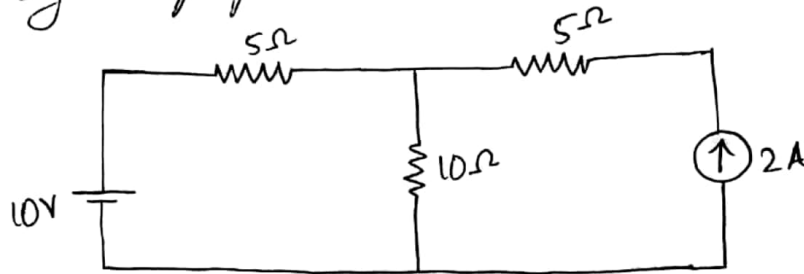


$$I_1 = I_1' - I_1''$$

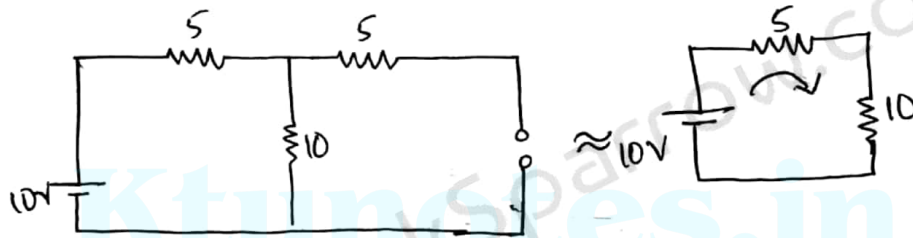
$$I_2 = I_2' - I_2''$$

$$I_3 = I_3' - I_3''$$

Q. Find the current through 10Ω resistance by superposition theorem.

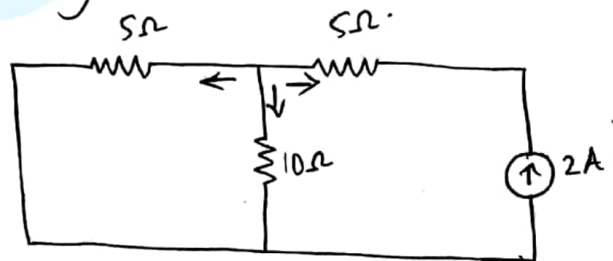


Solⁿ:



$$\therefore R = 15\Omega \quad V = 10V \quad \therefore I = \frac{10}{15} = 0.667A \downarrow$$

Considering current source,



$$\frac{V}{5} + \frac{V}{10} - 2 = 0 \Rightarrow V = \frac{20}{3}$$

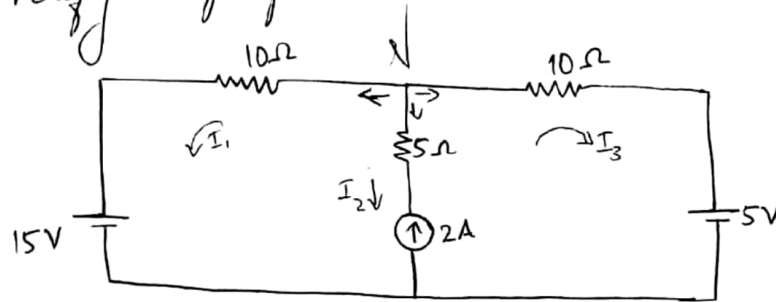
Current through 10Ω ,

$$\frac{V}{10} = \frac{20}{3 \cdot 10} = \underline{\underline{0.667 \downarrow}}$$

$$\therefore \text{total current} = 0.667 \text{ A} + 0.667 \text{ A}$$

$$\underline{\underline{I = 1.33 \text{ A} \downarrow}}$$

Q: Verify superposition theorem.



Solⁿ:
$$\frac{V-15}{10} - 2 + \frac{V-5}{10} = 0$$

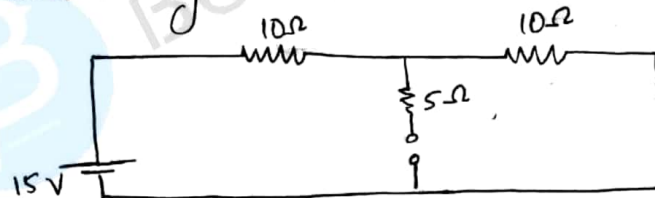
$$\Rightarrow \frac{V}{5} - 2 - 2 = 0 \Rightarrow \underline{\underline{V = 20}}$$

$$I_1 = \frac{V-15}{10} = \underline{\underline{0.5 \text{ A}}}$$

$$I_2 = \underline{\underline{-2 \text{ A} \downarrow}}$$

$$I_3 = \underline{\underline{1.5 \text{ A}}}$$

Considering 15V,

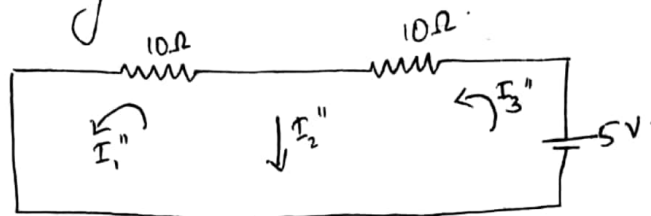


$$I_1' = \frac{15}{20} = \underline{\underline{0.75 \text{ A}}}$$

$$I_2' = \underline{\underline{0 \text{ A}}}$$

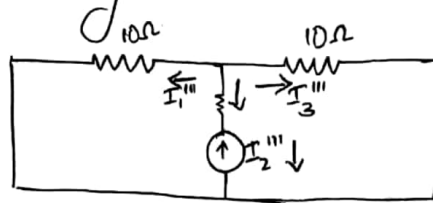
$$I_3' = \underline{\underline{0.75 \text{ A}}}$$

Considering 5V,



$$I_1'' = \frac{5}{20} = 0.25 \text{ A} \quad I_2'' = 0 \text{ A} \quad I_3'' = 0.25 \text{ A}$$

Considering current source,



By KCL, $\frac{V}{10} - 2 + \frac{V}{10} = 0 \Rightarrow \underline{V = 10\text{V}}$

$$I_1''' = 1 \text{ A} \quad I_2''' = -2 \text{ A} \downarrow \quad I_3''' = 1 \text{ A}$$

$$I_1 = -0.75 + 1 + 0.25 = \underline{0.5 \text{ A} \leftarrow}$$

$$I_2 = 0 + 0 - 2 = \underline{-2 \text{ A} \downarrow}$$

$$I_3 = 0.75 - 0.25 - 1 = \underline{1.5 \text{ A} \rightarrow}$$

$\therefore$ Theorem verified.

2) Thevenin's theorem:

It states that, in any bilateral network, consisting of any no. of voltage source, current source, resistance, impedance etc can be replaced by ~~the~~ a single

voltage source V_{th} in series with resistance R_{th} , where $V_{th} \rightarrow$ voltage across open ckted o/p terminal and $R_{th} \rightarrow$ effective resistance seen into network from the open ckted o/p terminal after making the effect of sources equal to zero by open ckting the current source and short ckting the voltage source if any.

Step 1:
Identify o/p terminal and open ckt it.

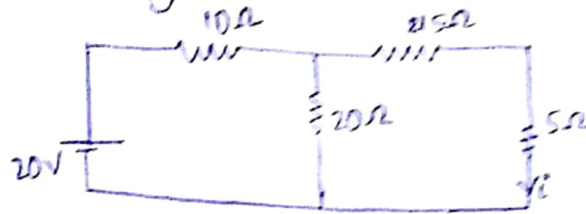
Step 2:
Find the voltage across the open ckted o/p terminal (V_{th})

Step 3:
Short ckt the voltage source and open ckt the current source

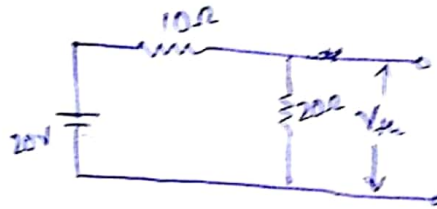
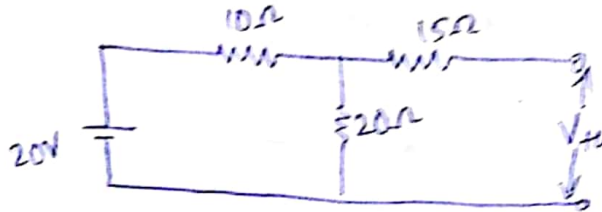
Step 4:
Find effective resistance R_{th} .

Step 5:
Place V_{th} in series with R_{th} and replace open ckted o/p terminal.

Q. Find I by Thevenin's theorem.



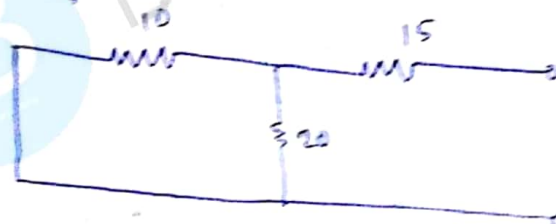
Solⁿ



$$V_{th} = \frac{20 \times 20}{10 + 20} = 13.3 \text{ V.}$$

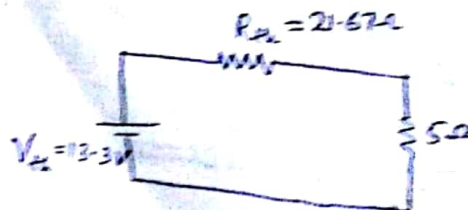
V_{th} is voltage across open Ω total Ω

To find R_{th}



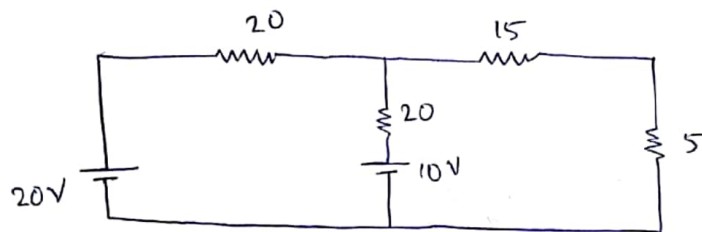
$$R_{th} = 10 // 20 + 15$$

$$= 21.6 \Omega$$



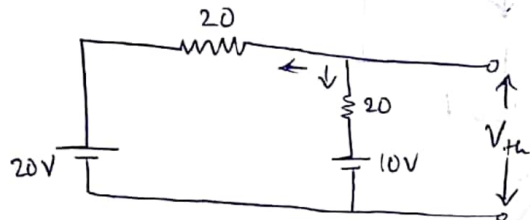
$$I = \frac{V}{R} = \frac{13.3}{5} = 2.66 \text{ A}$$

Q:



Calculate current through 5Ω resistance using
Thevenin's theorem.

Solⁿ:



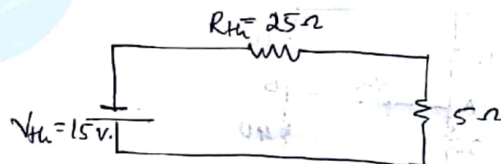
By node analysis, $\frac{V-20}{20} + \frac{V-10}{20} = 0$

$$\therefore V = 15V.$$

$$\underline{V_{th} = 15V.}$$

$$R_{th} = 20 \parallel 20 + 15$$

$$\underline{R_{th} = 25\Omega.}$$

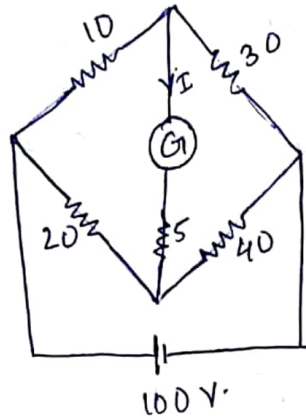


$$I = \frac{V}{R.}$$

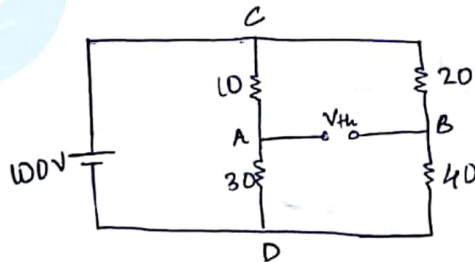
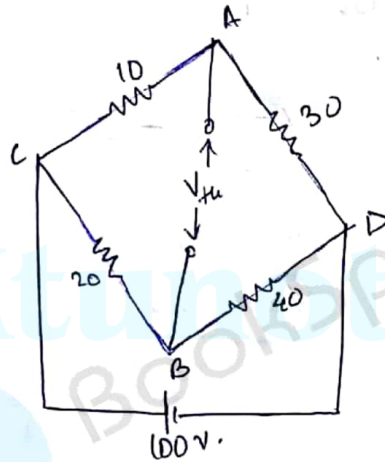
$$= \frac{15}{30}$$

$$= \underline{0.5A.}$$

Q: Find current through galvanometer using Thevenin's theorem.



Solⁿ:



$$V_{th} = V_A - V_B$$

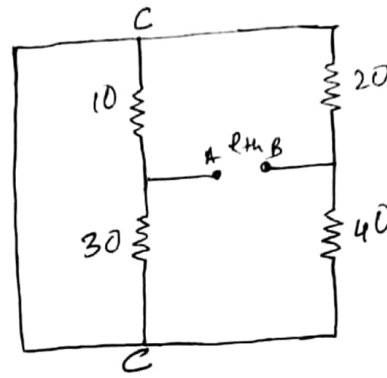
By voltage division rule,

$$V_A = \frac{100 \times 30}{10 + 30} = \underline{75V}$$

$$V_B = \frac{100 \times 40}{20 + 40} = \underline{66.6V}$$

$$\left. \begin{array}{l} V_A = 75V \\ V_B = 66.6V \end{array} \right\} V_{th} = \underline{8.33V}$$

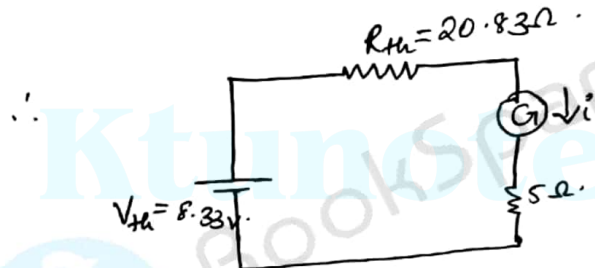
To find R_{th} ,



$$V_{AB} = V_A - V_B$$



$$R_{th} = 20.83$$



$$I = \frac{V}{R} = \frac{8.33}{20.83 + 5} = \underline{\underline{0.32 \text{ A}}}$$

4/8/16

3) Norton's Theorem:

It states that, in any linear bilateral network, consisting of any no. of voltage source, current source, resistance etc, can be ~~to~~ represented by a single current source I_N in $||^u$

with a resistance R_N , where I_N is the current across short ckt of terminal and R_N is the effective resistance seen into the network from the open ckt of terminal. After making the effects of sources equal to zero, by open cktg current source and short cktg the voltage source, if any.

Step 1:

Identify o/p terminal and short ckt it.

Step 2:

Find current across short ckted o/p terminal. (I_N)

Step 3:

Short ckt the voltage source and open ckt the current source.

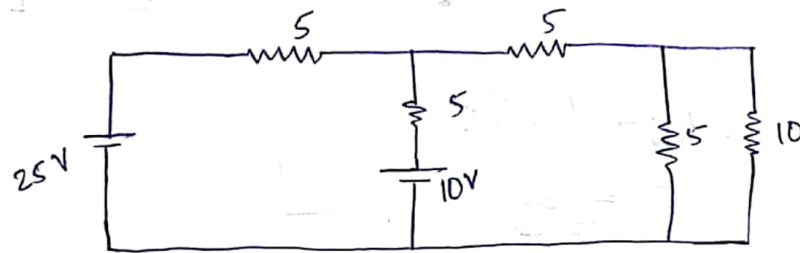
Step 4:

Find the effective resistance R_N .

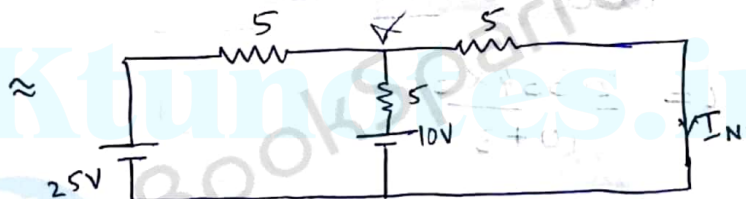
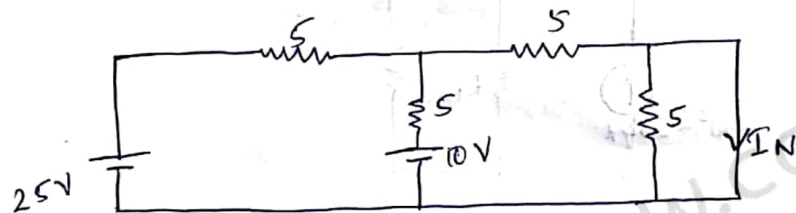
Step 5:

Place $I_N \parallel R_N$ and replace the short o/p terminal.

Q: Determine current through 10Ω resistance by Norton's theorem.



Solⁿ:



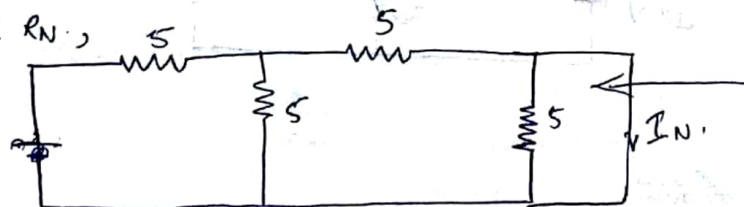
$$\frac{V-25}{5} + \frac{V-10}{5} + \frac{V}{5} = 0$$

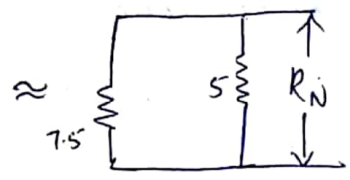
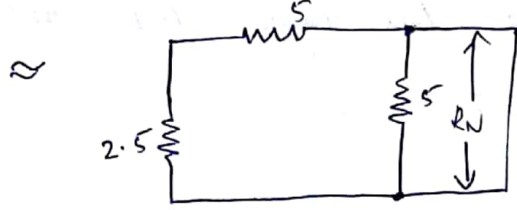
$$\frac{3V}{5} - 35 = 0 \Rightarrow \frac{3V}{5} = 35 \Rightarrow 3V = 175 \Rightarrow V = \frac{175}{3} = 58.33$$

$$\Rightarrow V = 11.67 \text{ V}$$

$$I_N = \frac{V}{5} = \frac{11.67}{5} = 2.334 \text{ A}$$

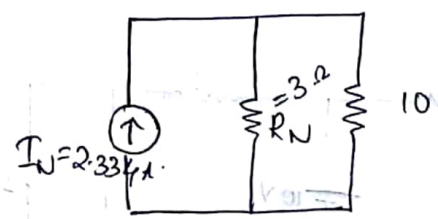
To find R_N ,





$$\therefore R_N = 7.5 // 5$$

$$= 3 \Omega$$

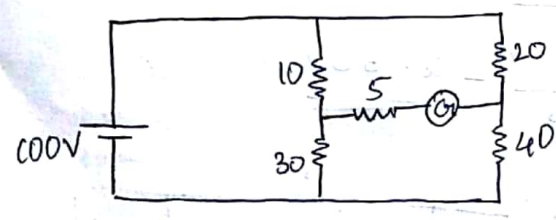


By current division rule,

$$i = \frac{2.334 \times 3}{10 + 3}$$

$$i = 0.53 \text{ A}$$

HW
Q: 1) Find current through galvanometer by Norton's theorem.

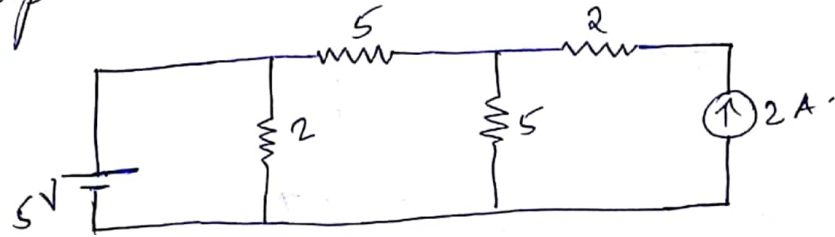


$$R_N = 20.83 \Omega$$

$$I_N = 0.401 \text{ A}$$

$$i_G = 0.32 \text{ A}$$

Q2) Using superposition theorem, find the power dissipated in the 5Ω resistance.

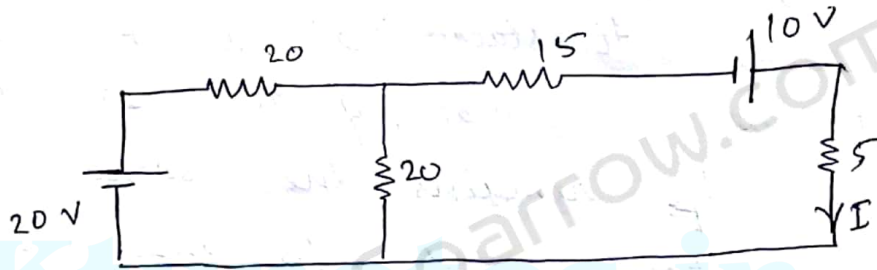


$$P = I^2 R$$

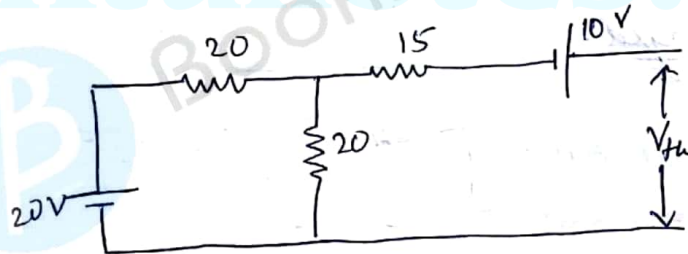
$$P = \frac{V^2}{R}$$

$$P = VI$$

Q: Find the current by Thevenin's theorem



Sol:

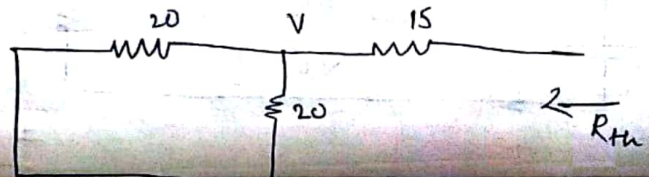


By node analysis,

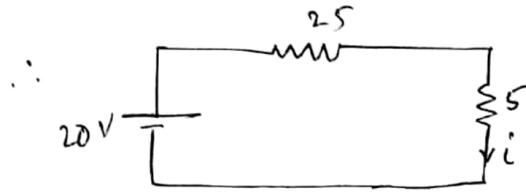
$$\frac{V - 20}{20} + \frac{V}{20} = 0 \Rightarrow V = 10V$$

$$V_{th} = V + 10 = 10 + 10 = \underline{\underline{20V}}$$

To find R_{th}



$$\therefore R_{th} = 25 \Omega$$

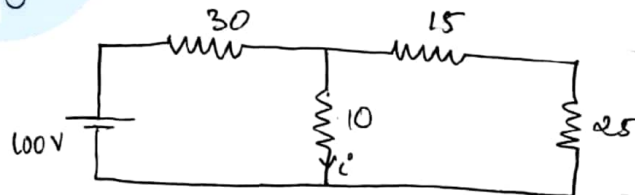


$$I = \frac{20}{30} = \frac{2}{3} = 0.667A$$

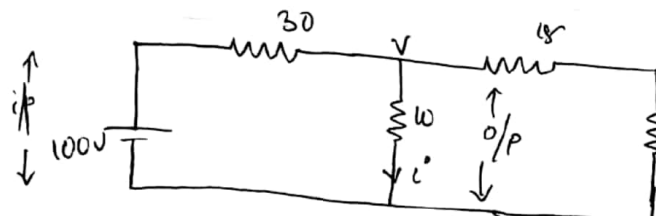
4) Reciprocity Theorem:

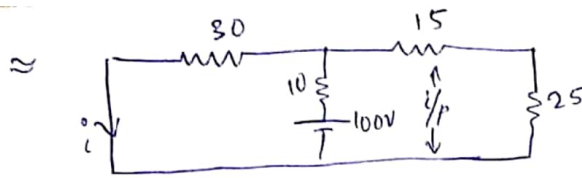
It states that, a ckt is said to be reciprocal, if the ratio of o/p to i/p remains the same even if the position of o/p and i/p are interchanged.

Q: Verify reciprocity theorem.



Solⁿ:





[Inteuchaged]

$$\frac{V-100}{30} + \frac{V}{10} + \frac{V}{25+15} = 0$$

$$\frac{4V}{30} - \frac{100}{30} + \frac{V}{40} = 0$$

$$\Rightarrow \underline{V = 21.05 \text{ V}}$$

Before
intteuchaged.

$$i = \frac{V}{10} = \frac{21.05}{10} = \underline{2.105 \text{ A}}$$

Solving, intteuchaged ckt.

$$\frac{V}{30} + \frac{V-100}{10} + \frac{V}{25+15} = 0$$

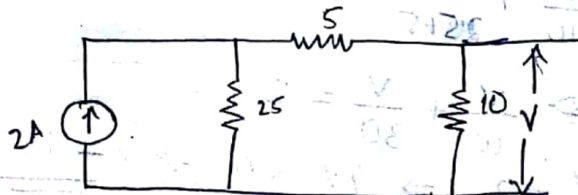
$$\frac{V}{30} - \frac{V}{10} - 10 + \frac{V}{40} = 0$$

$$\frac{4V}{120} - \frac{120}{120} + \frac{3V}{120} = 10$$

$$\Rightarrow \underline{V = 63.1579 \text{ V}}$$

$$i = \frac{V}{30} = \frac{63.1579}{30} = \underline{2.105 \text{ A}}$$

Q:



Solⁿ

By node analysis,

$$-2 + \frac{V}{25} + \frac{V}{10+5} = 0$$

$$\frac{V}{25} + \frac{V}{15} = -2$$

$$\Rightarrow V = \underline{\underline{18.75 \text{ V}}}$$

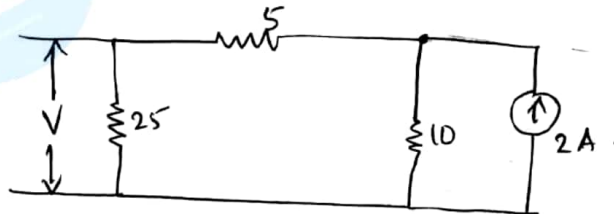
$$i = \frac{V}{10+5} = \frac{18.75}{15} =$$

By voltage division rule,

$$V_{10} = \frac{18.75 \times 10}{15}$$

$$V_{10} = \underline{\underline{12.5 \text{ V}}}$$

By interchanging,



By node analysis

$$-2 + \frac{V}{10} + \frac{V}{25+5} = 0$$

$$\frac{V}{10} + \frac{V}{30} = 2$$

$$\frac{4V}{30} = 2 \Rightarrow 4V = 60 \Rightarrow V = \underline{\underline{15 \text{ V}}}$$

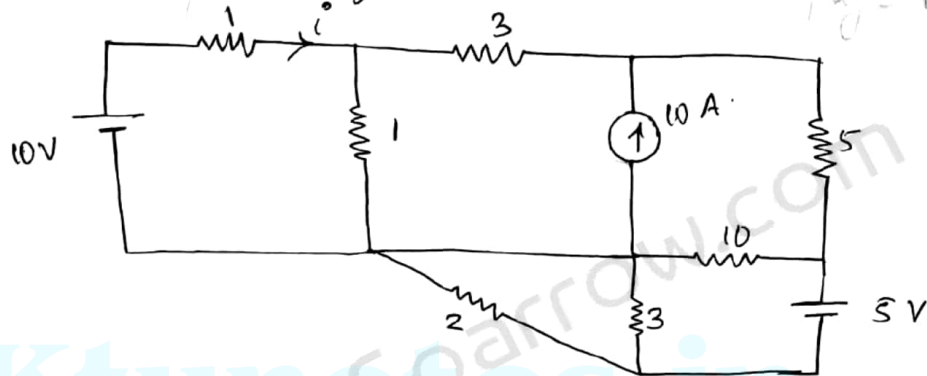
V_{25} by voltage div. rule,

$$V_{25} = \frac{15 \times 25}{30}$$

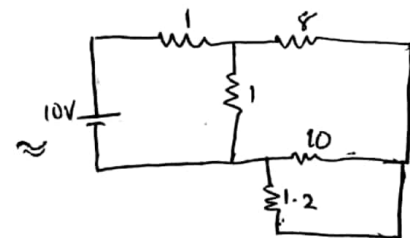
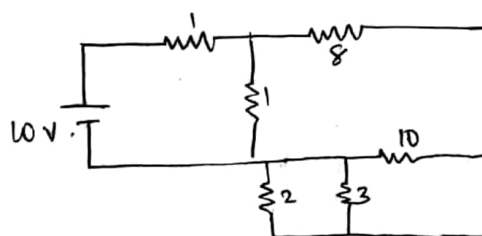
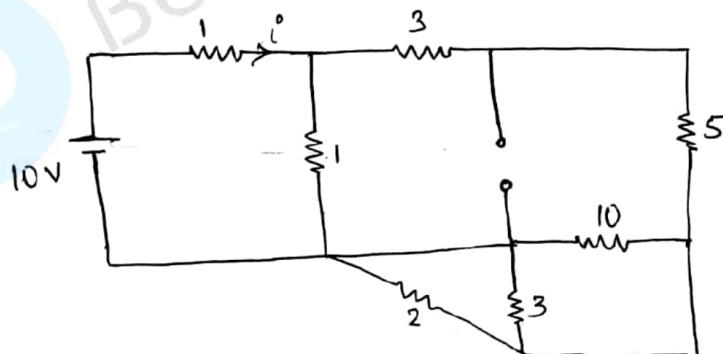
$$\underline{\underline{V_{25} = 12.5 \text{ V}}}$$

Q: Find i by superposition theorem.

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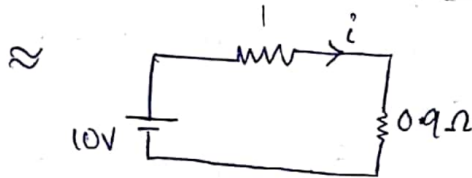
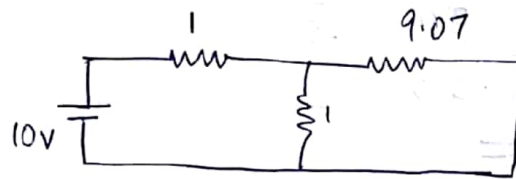


Solⁿ: Considering 10V,



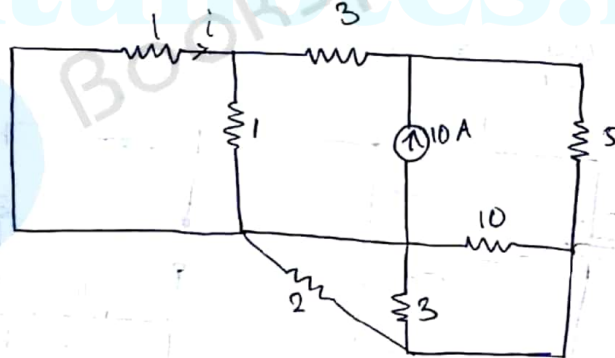
$$\cancel{1.07} \parallel 1.07 \text{ with } 8 = 9.07$$

$$1 \parallel 9.07 = 0.9 \Omega$$



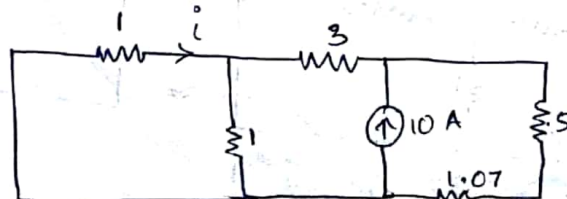
$$i = \frac{V}{R} = \frac{10}{1.09} = \underline{\underline{5.26 \text{ A}}}$$

Considering 10 A source,

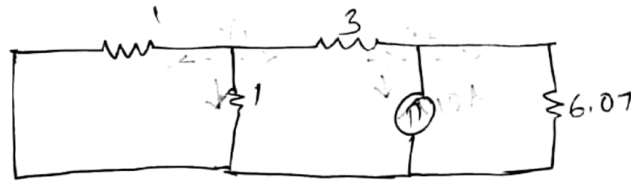


$$2 \text{ and } 2 \parallel 3 = 1.2$$

$$1.2 \parallel 10 = 1.07$$



$$R_{\text{total}} = 6.07$$



$$\frac{V_1}{1} + \frac{V_1}{1} + \frac{V_1 - V_2}{3} = 0$$

$$2V_1 + \frac{V_1}{3} - \frac{V_2}{3} = 0$$

$$\frac{7V_1}{3} - \frac{V_2}{3} = 0 \Rightarrow 7V_1 - V_2 = 0 \quad \text{--- (1)}$$

$$\frac{V_2 - V_1}{3} - 10 + \frac{V_2}{6.07} = 0$$

$$\frac{V_2}{3} + \frac{V_2}{6.07} - \frac{V_1}{3} = 10$$

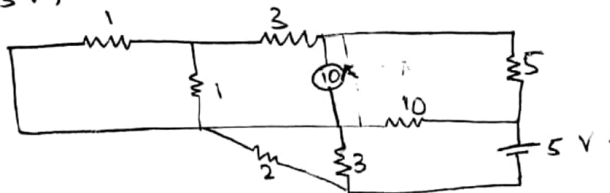
$$\frac{6.07}{18.21} 2V_2 + 3V_2 - \frac{V_1}{3} = 10 \Rightarrow \frac{9.07V_2}{18.21} - \frac{V_1}{3} = 10$$

$$\frac{V_2}{2} - \frac{V_1}{3} = 10 \Rightarrow 3V_2 - 2V_1 = 60 \quad \text{--- (2)}$$

From (1) and (2),

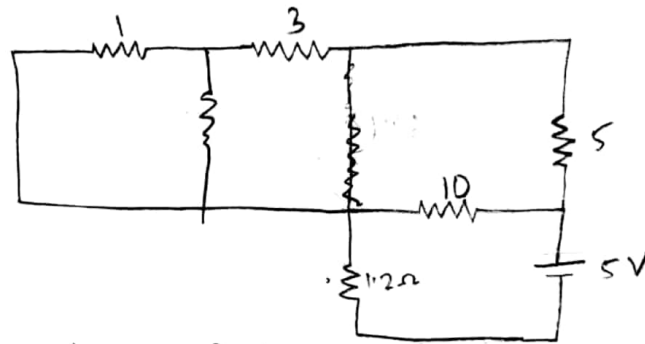
$$\left. \begin{array}{l} 2V_1 - 3V_2 = 0 \\ -2V_1 + 3V_2 = 60 \end{array} \right\} \Rightarrow 19V_1 = 60 \Rightarrow V_1 = \underline{\underline{3.15V}}$$

$\therefore$ Current through $1\Omega = \frac{V_1}{1} = \underline{\underline{3.15A}}$
Consider SV,

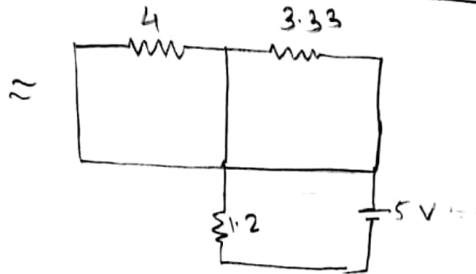
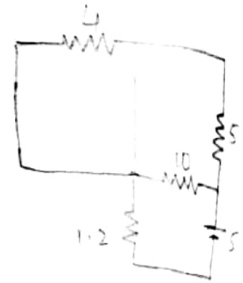


$$1\Omega // 1\Omega = 2\Omega$$

$$2\Omega // 3\Omega = 1.2\Omega$$



~



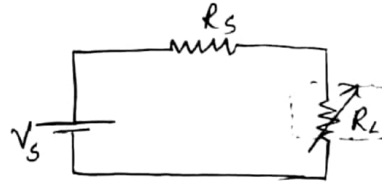
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5/4/16

Maximum power transfer theorem: dc



It states that, max. power will be transferred from the source to the load, only when $R_L = R_s$, where load resistance is equal to source resistance.

In order to maximise power,

$$\frac{dP}{dR_L} = 0 \quad \text{--- (1)}$$

$$\text{Power} = \frac{V^2}{R}$$

$$\text{We know } I = \frac{V_s}{R_s + R_L}$$

$$P = \frac{V_s^2}{R_s + R_L}$$

$$P = I^2 R_L$$

$$\Rightarrow P = \frac{V_s^2}{(R_s + R_L)^2} \cdot R_L$$

$$\therefore \text{(1)} \Rightarrow \frac{d}{dR_L} \left[\frac{V_s^2 \cdot R_L}{(R_s + R_L)^2} \right] = 0$$

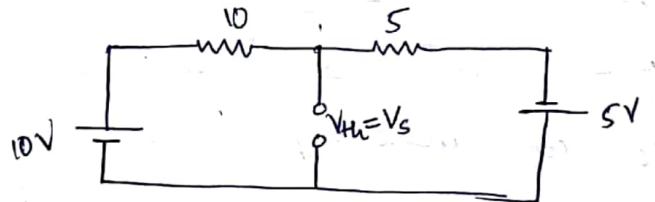
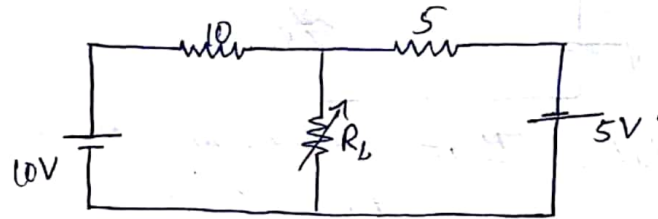
$$\Rightarrow (R_s + R_L)^2 \cdot V_s^2 - 2V_s^2 R_L (R_s + R_L) = 0$$

$$\Rightarrow (R_s^2 + 2R_s R_L + R_L^2) V_s^2 - 2V_s^2 R_s R_L + 2V_s^2 R_L^2 = 0$$

$$\Rightarrow \cancel{V_s^2 R_s^2} + 2\cancel{R_s R_L V_s^2} + \cancel{R_L^2 V_s^2} - 2\cancel{V_s^2 R_s R_L} + 2\cancel{V_s^2 R_L^2} = 0$$

$$\Rightarrow \underline{\underline{R_s^2 - R_L^2}} \quad R_s = R_L$$

Q: Determine the value of R_L for max power and hence calculate the max. power.



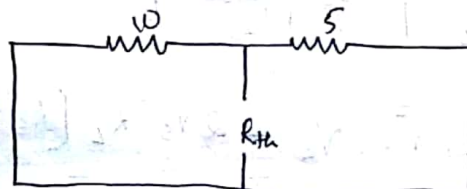
$$\frac{V-10}{10} + \frac{V+5}{5} = 0 \Rightarrow \underline{V=0}$$

$$\frac{3V}{10} - 12 = 0 \Rightarrow 3V = 120$$

$$\Rightarrow 3V = 120$$

ie $\underline{V_{th} = 0}$

To find R_{th}



$$\underline{10 \parallel 5 = 3.33 \Omega}$$

$$R_{th} = 3.33 \Omega$$

$$\underline{R_{th} = R_L}$$

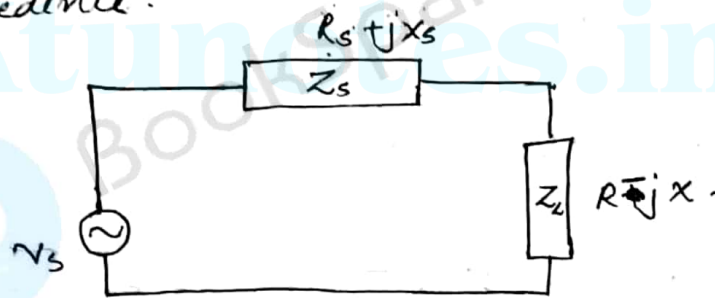


$$P = I^2 R$$

$$= 0$$

Max power transfer theorem for AC net

It states that max. power will be transferred from source to the load, when the load impedance is the complex conjugate of the source impedance.



Condition for max. power transfer;

$$R_s = R$$

$$X_s = -X$$

$$P = I^2 R$$

$$I = \frac{V_s}{R_s + R + j(X_s + X)}$$

$$\therefore \text{Magnitude of } I = \frac{V_s}{\sqrt{(R_s + R)^2 + (X_s + X)^2}}$$

$$\therefore P = I^2 R$$

$$= \frac{V^2 R}{(R_s + R)^2 + (X_s + X)^2}$$

In order to maximise power,

$$\frac{dP}{dR} = 0 \text{ should give } R_s = R$$

$$\text{and } \frac{dP}{dX} = 0 \text{ should give } X_s = -X$$

$$\frac{dP}{dR} = \frac{d}{dR} \left[\frac{V_s^2 R}{(R_s + R)^2 + (X_s + X)^2} \right] = 0$$

$$= \frac{(R_s + R)^2 + (X_s + X)^2 [V_s^2] -$$

$$= \left[(R_s + R)^2 + (X_s + X)^2 \right] (V_s^2) - V_s^2 R [2(R_s + R) + 2(X_s + X)] = 0$$

$$= (R_s + R)^2 + (X_s + X)^2 - R [2(R_s + R) + 2(X_s + X)] = 0$$

$$= R_s + R + (X_s + X)^2$$

$$= (R_s + R)^2 + (X_s + X)^2 = R [2(R_s + R)]$$

$$= (R_s + R)^2 = R [2(R_s + R)]$$

when $X_s = -X$.

$$= R_s + R = 2R$$

$$\Rightarrow \underline{R_s = R}$$

Now, $\frac{dP}{dx} = 0$

$$\Rightarrow \frac{d}{dx} \left[\frac{V_s^2 R}{(R_s + R)^2 + (X_s + X)^2} \right] = 0$$

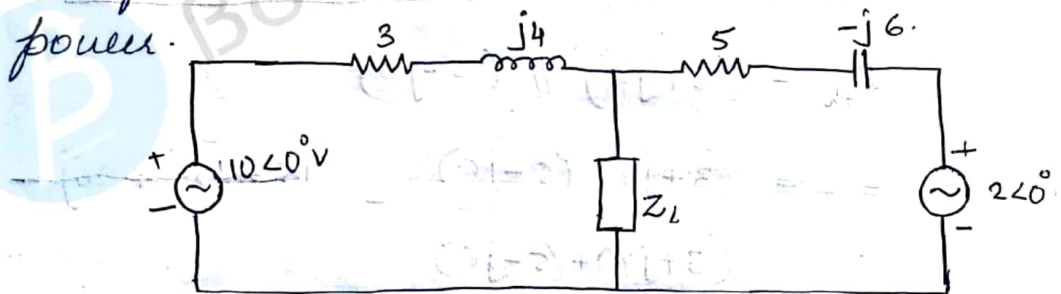
$$\Rightarrow (R_s + R)^2 + (X_s + X)^2 \cdot 0 - V_s^2 R \cdot 2(X_s + X) = 0$$

$$\Rightarrow -V_s^2 R \cdot 2(X_s + X) = 0$$

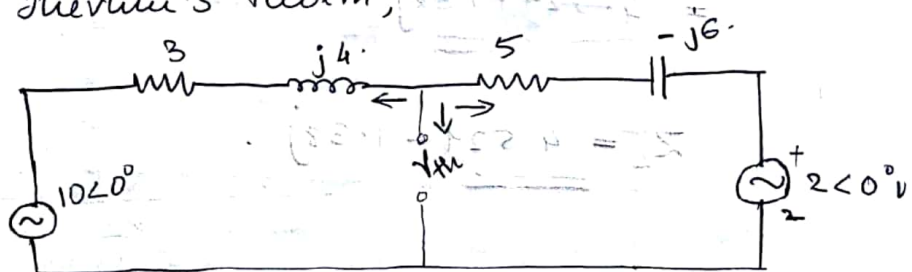
$$\Rightarrow X_s + X = 0$$

$$\Rightarrow \underline{X_s = -X}$$

Q: Determine value of Z_L for max. power transfer and hence calculate the max. power.



Sol^y: By Thevenin's theorem,



By node analysis.

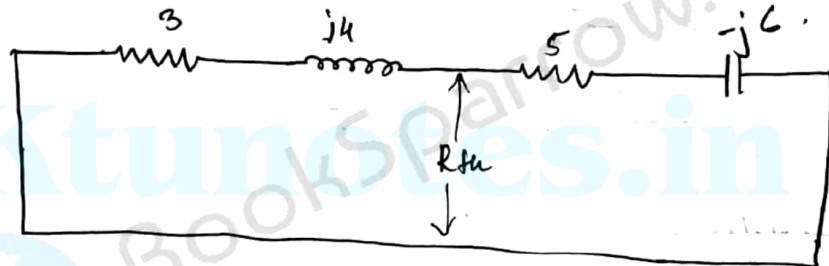
$$\frac{V - 10 \angle 0^\circ}{3 + j4} + \frac{V - 2 \angle 0^\circ}{5 - j6} = 0$$

$$(V - 10)(5 - j6) + (V - 2)(3 + j4) = 0$$

$$\Rightarrow 5V - 6Vj - 50 + 60j + 3V + 4Vj - 6 - 8j = 0$$

$$\Rightarrow 8V - 56 - 62j - 2Vj = 0$$

$$\Rightarrow \underline{V = 8.117 - 4.47j}$$



$$R_{th} = (3 + j4) \parallel (5 - j6)$$

$$= \frac{(3 + j4)(5 - j6)}{(3 + j4) + (5 - j6)} = \frac{15 - 18j + 20j - 24}{8 - 2j}$$

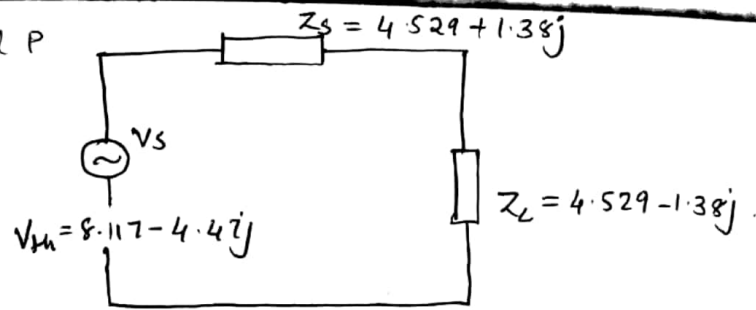
$$= \underline{4.529 + 1.38j}$$

$$\therefore \underline{Z_L = 4.529 - 1.38j}$$

$$\frac{15 - 18j + 20j - 24}{8 - 2j}$$

$$\begin{aligned} &= \frac{39 + 2j}{8 - 2j} \\ &= \frac{(39 + 2j)(8 + 2j)}{(8 - 2j)(8 + 2j)} \\ &= \frac{312 + 78j + 16j - 4}{64 - 16j^2} \\ &= \frac{308 + 94j}{80} \end{aligned}$$

To find P

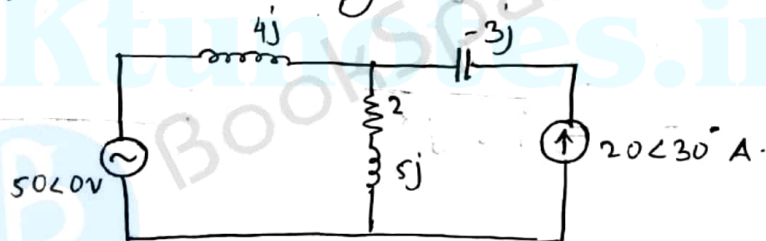


$$P = I^2 R = \left(\frac{8.117 - 4.47j}{4.529 + 4.529} \right) \times 4.529$$

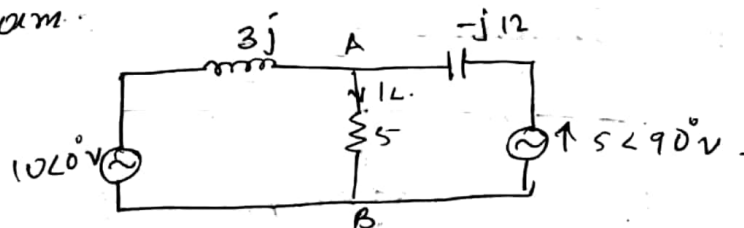
$$= \underline{\underline{2.533 - 4.005j}}$$

HW

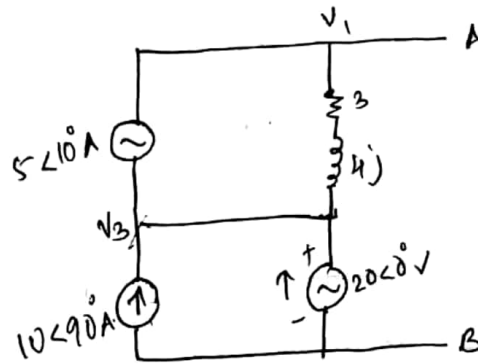
Q: Determine the voltage across $2 + 5j \Omega$ impedance using superposition theorem.



Q: Determine the load current I_L by Norton's theorem.



Q: Convert the ckt by a single voltage source in series with impedance.



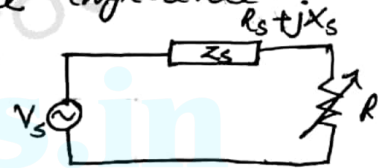
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If the load is

In ac analysis, if load is resistive, max power will be transferred from source to load when load resistance is equal to the magnitude of source impedance.

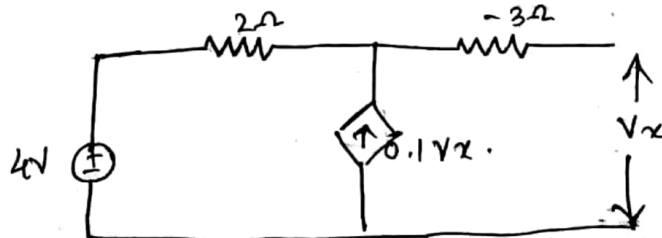
For max power,

$$R = \sqrt{R_s^2 + X_s^2}$$



Dependent Source.

Q: Obtain Thevenin's equivalent ckt.



$$V_{th} = V_{oc}$$

$$I_{sc} = I_N$$

Solⁿ:

$$R_{th} \text{ or } R_N = \frac{V_{oc}}{I_{sc}}$$

if dependent source is present in the ckt

Here $V_{oc} = V_{th}$

$i_{sc} = i_N$

$\therefore R_{th} \text{ or } R_N = \frac{V_{th}}{i_N}$

o/p terminal : V_x

$V_{oc} = V_x$

By node analysis

$\frac{V_x - 4}{2} - 0.1V_x = 0$

$\Rightarrow \underline{V_x = 5V}$

$\therefore \underline{V_{oc} = 5V}$

$i_{sc} = ?$

By node analysis,

$\frac{V - 4}{2} - 0.1V_x + \frac{V}{3} = 0$

$V_x = 0$

$\frac{V - 4}{2} + \frac{V}{3} = 0$

$\Rightarrow \frac{V}{2} + \frac{V}{3} = 4$

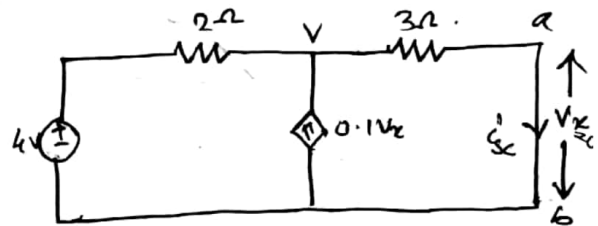
$\frac{5V}{6} = 4$

$\Rightarrow 5V = 24$

$V = 4.8$

$\therefore \underline{V = 2.4}$

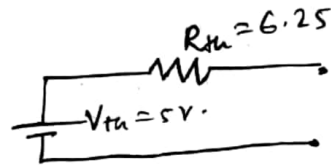
$\therefore i_{sc} = \frac{V}{3} = \frac{2.4}{3} \Rightarrow \underline{0.8A}$



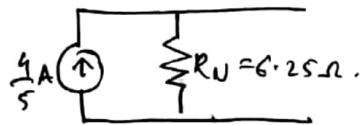
$\frac{1-0.1}{2} = \frac{1-0.2}{2} = 1-0$

$$R_{th} = \frac{V_{oc}}{i_s} = \frac{5}{4/5} = \underline{\underline{6.25 \Omega}}$$

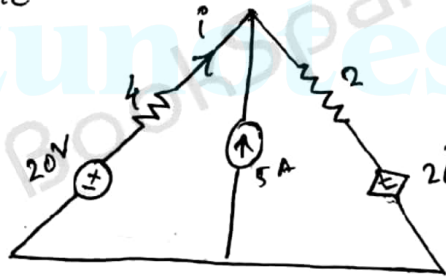
By Thevenin's theorem,
series connection.



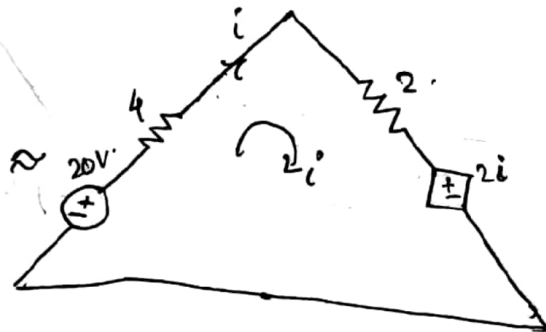
By Norton's theorem,



Q: Find the current i by superposition principle.



Sol: By consider 20V,



$$20 - 4i - 2i - 2i = 0$$

$$\Rightarrow i_1 = 2.5 \text{ A} \uparrow$$

Considering 5 A,



By node analysis,

$$\frac{V}{4} - 5 + \frac{V - 2i}{2} = 0$$

i is the current flowing through 4 Ω and V .

$$\therefore i = -\frac{V}{4} \text{ [opp direction]},$$

$$\therefore \frac{V}{4} - 5 + \frac{V + \frac{2V}{4}}{2} = 0$$

$$\Rightarrow \frac{V}{2} - 10 + V + \frac{2V}{4} = 0$$

$$\Rightarrow 2V = 10$$

$$\underline{\underline{V = 5 \text{ V.}}}$$

$$\therefore i = -\frac{V}{4}$$

$$i_2 = -\frac{5}{4} = \underline{\underline{-1.25 \text{ A.}}}$$

$$I = i_1 + i_2$$

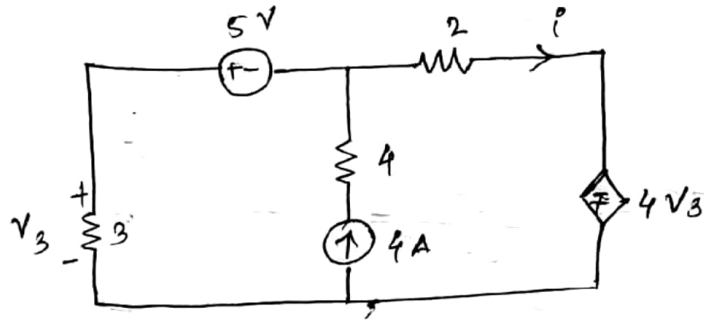
$$= 2.5 + (-1.25)$$

$$= \underline{\underline{1.25 \text{ A.}}}$$

$$\begin{array}{r} 2.5 \\ + (-1.25) \\ \hline 1.25 \end{array}$$

Q: Determine current through 2Ω resistance using superposition theorem.

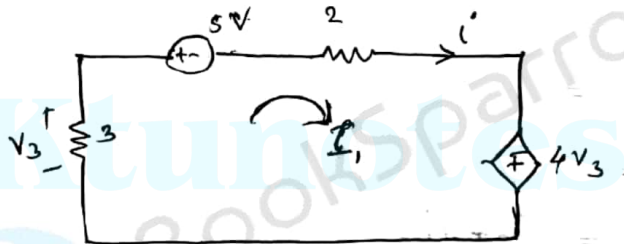
Sol:



Sol:

$4V_3 \rightarrow$ dependent source.

Consider 5V,



By mesh analysis,

$$-3I_1 - 5 + 2I_1 + 4V_3 = 0$$

$$8V_3 - 5 = 2I_1$$

$$\Rightarrow 8 - 9I_1 - 5 = 2I_1$$

$$\Rightarrow -9I_1 - 2I_1 = -3$$

$$\Rightarrow -11I_1 = -3$$

$$I_1 = \frac{-3}{-11} \Rightarrow I_1 = \frac{3}{11}$$

$$V_3 = ?$$

$$I_1 = ?$$

$$I_1 = \frac{V_3}{R}$$

$$I_1 = \frac{V_3}{3}$$

$$V_3 = -3I_1$$

$$-5I_1 - 5 + 4V_3 = 0$$

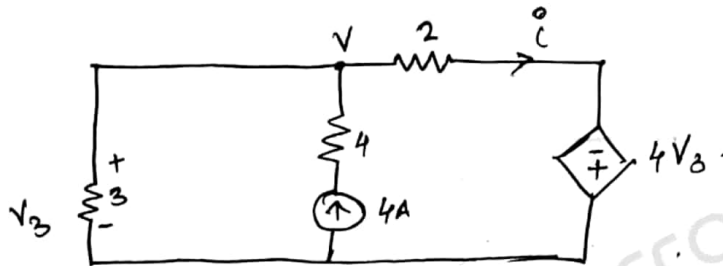
$$-5I_1 + 4 \cdot 3I_1 = 5$$

$$-5I_1 - 12I_1 = 5$$

$$-17I_1 = 5$$

$$I_1 = -0.294 \text{ A}$$

Consider 4A,



Here $V = V_3$.

$$+\frac{V_3}{3} - 4 + 4V_3 = 0$$

$$3V_3 = 4 \Rightarrow V_3 = \frac{4}{3} = 1.33$$

$$4V_3 = 4(1.33) = 5.32 \text{ A}$$

$$\frac{V_3}{3} - 4 + \frac{V_3 + 4V_3}{2} = 0$$

$$\frac{V_3}{3} + \frac{V_3}{2} + 2V_3 = 4 \Rightarrow \frac{5V_3}{6} + 2V_3 = 4$$

$$\Rightarrow 5V_3 + 12V_3 = 24$$

$$= 17V_3 = 24$$

$$V_3 = 3.4$$

$$\Rightarrow V_3 = 1.41 \text{ V}$$

$$\frac{V_3 + 4V_3}{2} = \frac{(1.41) + 4(1.41)}{2}$$

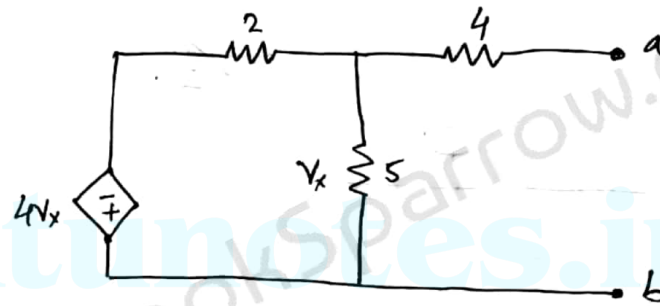
$$I_2 = \underline{\underline{3.52 \text{ A}}}$$

$$\therefore I = I_1 + I_2$$

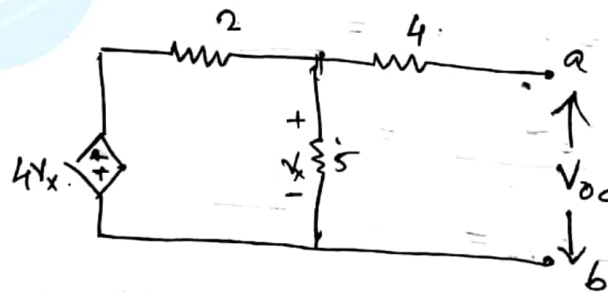
$$= -0.29 + 3.52$$

$$= \underline{\underline{3.23 \text{ A}}}$$

Q: Find Norton's equivalent ckt.



$$\text{Resistance} = \frac{V_{oc}}{I_{sc}}$$

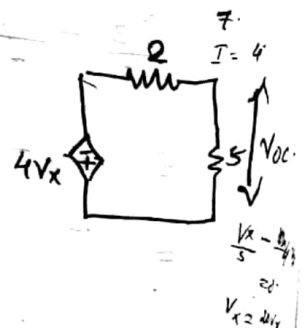


$I_{sc} \rightarrow$ current across short circuit of terminals a and b.

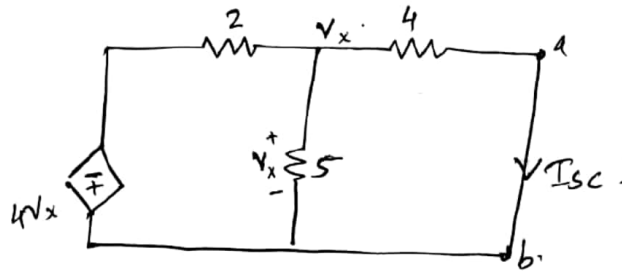
$$V_{oc} = V_x$$

$$\frac{V_x + 4V_x}{2} + \frac{V_x}{5} = 0$$

$$\Rightarrow \underline{\underline{V_{oc} = V_x = 0}}$$



$$I_{sc} = 0$$



$$\frac{V_x + 4V_x}{2} + \frac{V_x}{5} + I_{sc} = 0$$

$$\Rightarrow \underline{I_{sc} = 0}$$

As the ckt is having only dependent source, $V_{oc} = I_{sc} = 0$

Apply a 1A source externally and determine the resultant voltage across it, and then find $R_{th} = \frac{V}{I} = \frac{V}{1}$

or

1V source externally and determine current through it and then find,

$$R_{th} = \frac{1}{I}$$

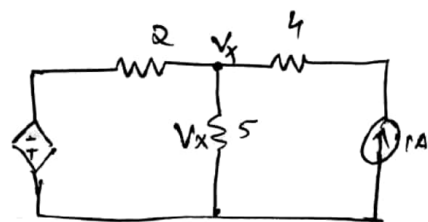
By node analysis

$$\frac{V_x + 4V_x}{2} + \frac{V_x}{5} - 1 = 0$$

$$\Rightarrow \frac{5V_x}{2} + \frac{V_x}{5} = 1 \Rightarrow 25V_x + 2V_x = 10$$

$$27V_x = 10$$

$$\underline{V_x = 10/27 = 0.37 V}$$



$$\begin{aligned}
 V_x \text{ across } 1A &= \text{Total voltage} - \text{voltage across } 4\Omega \\
 &= 0.37 - [-1 \times 4] \\
 &= 0.37 + 4 = \underline{\underline{4.37}}
 \end{aligned}$$

$$R_{th} = ?$$

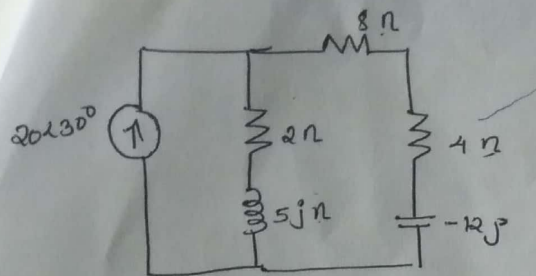
$$R = \frac{V}{I}$$

$$R_{th} = \frac{4.37}{1} = \underline{\underline{4.37\Omega}}$$

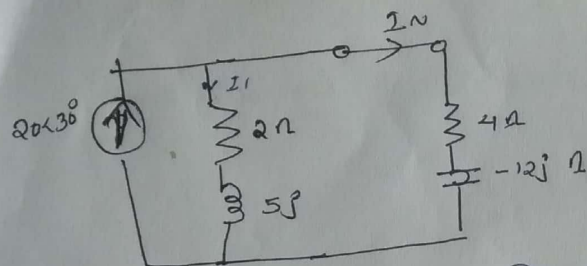
$\therefore$ Resultant Norton's ckt :



arkinnoy.com



Find the voltage across 8Ω resistance & also power dissipated across 8Ω resistance using Norton's Theorem.

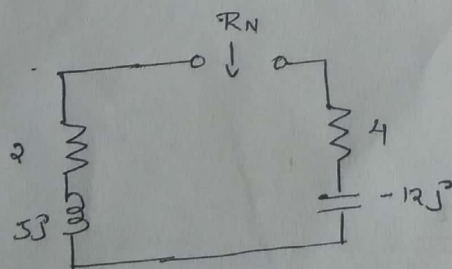


By Current Division

$$I_N = \frac{20\angle 30^\circ \times (2 + 5j)}{2 + 5j + 4 + -12j}$$

$$= \frac{(20\angle 30^\circ)(2 + 5j)}{(6 - 7j)} = 11.68 \angle 147.5^\circ$$

R_N



$$R_N = 2 + 5j + 4 + -12j$$

$$= 6 - 7j$$

$$= 9.22 \angle -49.39^\circ$$

By KCL

$$I = I_1 + I_N$$

$$20\angle 30^\circ = \frac{V}{2 + 5j} + \frac{V}{4 - 12j}$$

$$= V \left[\frac{1}{2 + 5j} + \frac{1}{4 - 12j} \right]$$

$$20\angle 30^\circ = V [0.135 \angle -46.03^\circ]$$

$$V = \frac{20\angle 30^\circ}{0.135 \angle -46.03^\circ}$$

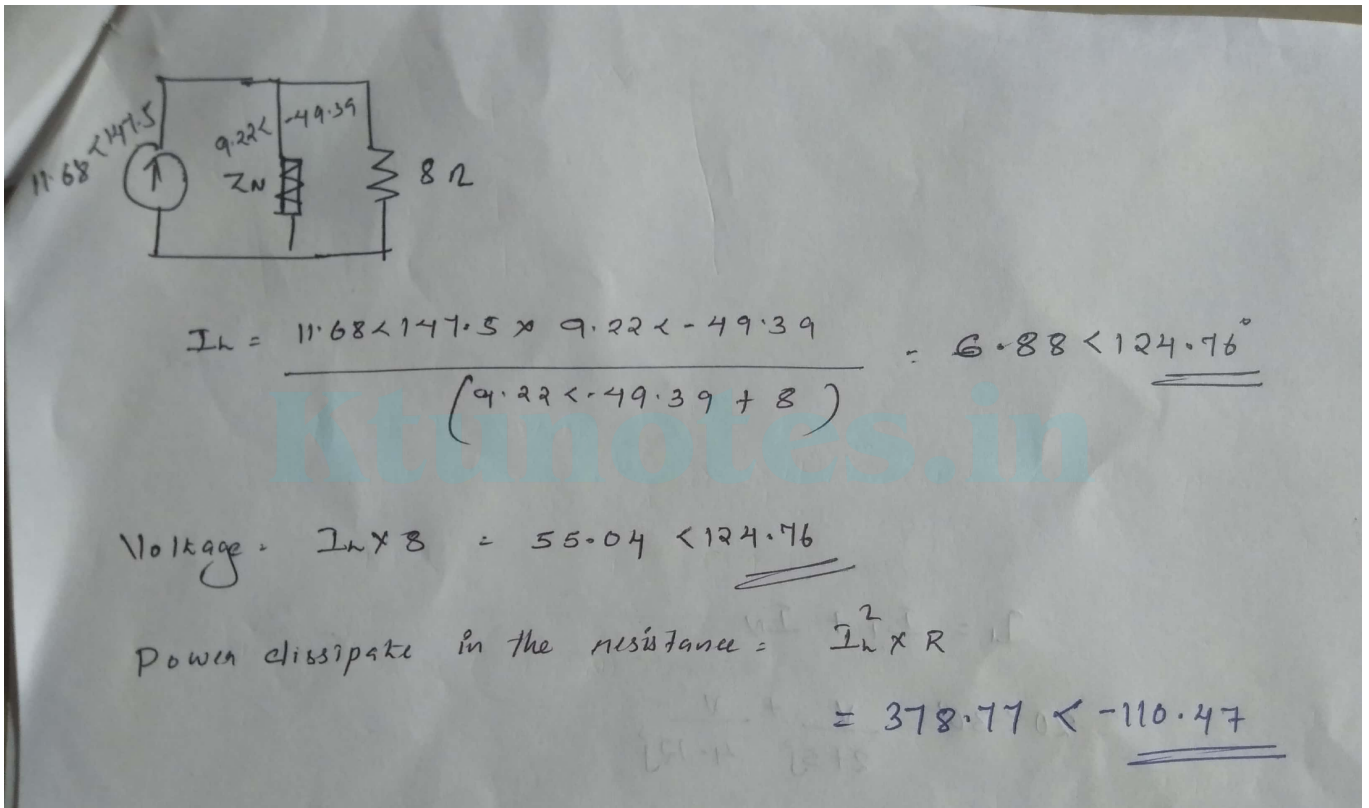
$$V = 147.76 \angle 76.03^\circ$$

$$I_N = \frac{V}{4 - 12j}$$

$$I_N = 11.68 \angle 147.59^\circ$$

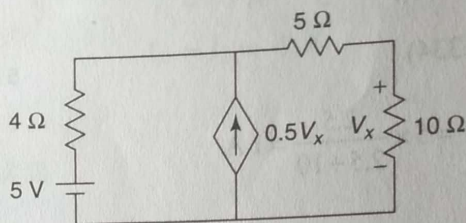
R

Shot on realme 2



Example 3.77

Find the current through the $10\ \Omega$ resistor for the network of Fig. 3.339.

**Fig. 3.339****Solution**

Step I Calculation of V_{Th} (Fig. 3.340)

For the mesh,

$$I = -0.5V_x = -0.5V_{Th}$$

Writing the V_{Th} equation,

$$5 - 4I - 0 - V_{Th} = 0$$

$$5 - 4(-0.5V_{Th}) - V_{Th} = 0$$

$$V_{Th} = -5\text{ V}$$

Step II Calculation of I_N (Fig. 3.341)

From Fig. 3.341,

$$V_x = 0$$

The dependent source of $0.5V_x$ depends on the controlling variable V_x . When $V_x = 0$, the dependent source vanishes, i.e. $0.5V_x = 0$ as shown in Fig. 3.342.

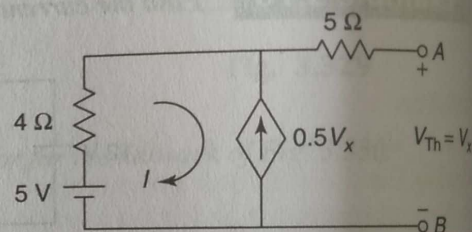
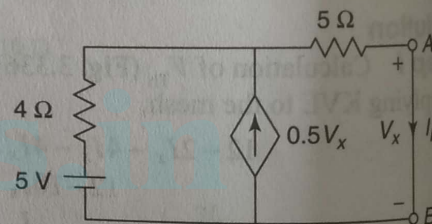
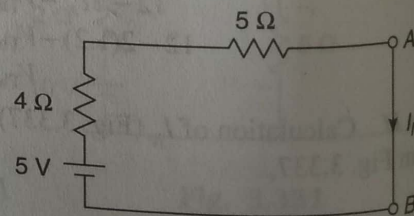
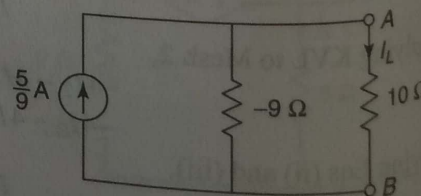
$$I_N = \frac{5}{4+5} = \frac{5}{9}\text{ A}$$

Step III Calculation of R_N

$$R_N = \frac{V_{Th}}{I_N} = \frac{-5}{\frac{5}{9}} = -9\ \Omega$$

Step IV Calculation of I_L (Fig. 3.343)

$$I_L = \frac{5}{9} \times \frac{-9}{-9+10} = -5\text{ A}$$

**Fig. 3.340****Fig. 3.341****Fig. 3.342****Fig. 3.343**

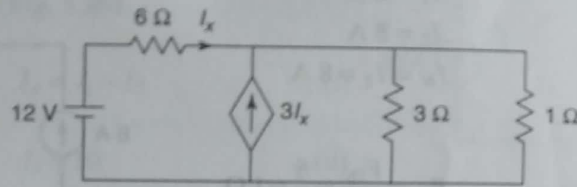
Example 3.80Find the current through the $1\ \Omega$ resistor in the network of Fig. 3.353.

Fig. 3.353

Solution**Step I** Calculation of V_{Th} (Fig. 3.354)

From Fig. 3.354,

$$I_x = I_1 \quad \dots(i)$$

Meshes 1 and 2 will form a supermesh.

Writing the current equation for the supermesh,

$$I_2 - I_1 = 3I_x = 3I_1$$

$$4I_1 - I_2 = 0 \quad \dots(ii)$$

Applying KVL to the outer path of the supermesh,

$$12 - 6I_1 - 3I_2 = 0$$

$$6I_1 + 3I_2 = 12 \quad \dots(iii)$$

Solving Eqs (ii) and (iii),

$$I_1 = 0.67\text{ A}$$

$$I_2 = 2.67\text{ A}$$

Writing the V_{Th} equation,

$$3I_2 - V_{Th} = 0$$

$$3(2.67) - V_{Th} = 0$$

$$V_{Th} = 8\text{ V}$$

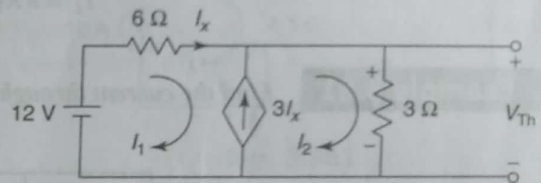


Fig. 3.354

Step II Calculation of I_N (Fig. 3.355)When a short circuit is placed across a $3\ \Omega$ resistor, it gets shorted as shown in Fig. 3.356.

From Fig. 3.356,

$$I_x = I_1 \quad \dots(i)$$

Meshes 1 and 2 will form a supermesh.

Writing the current equation for the supermesh,

$$I_2 - I_1 = 3I_x = 3I_1$$

$$4I_1 - I_2 = 0 \quad \dots(ii)$$

Applying KVL to the outer path of the supermesh,

$$12 - 6I_1 = 0$$

$$I_1 = 2 \quad \dots(iii)$$

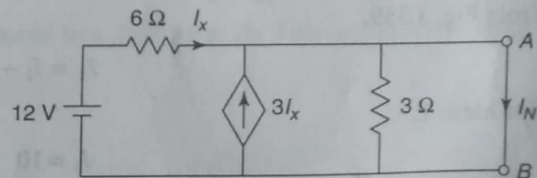


Fig. 3.355

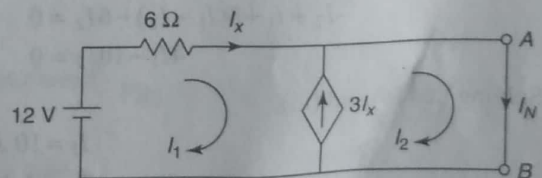


Fig. 3.356

R

Shot on realme 2

3.90 Network Analysis and Synthesis

Solving Eqs (ii) and (iii),

$$I_1 = 2 \text{ A}$$

$$I_2 = 8 \text{ A}$$

$$I_N = I_2 = 8 \text{ A}$$

Step III Calculation of R_N

$$R_N = \frac{V_{Th}}{I_N} = \frac{8}{8} = 1 \Omega$$

Step IV Calculation of I_L (Fig. 3.357)

$$I_L = 8 \times \frac{1}{1+1} = 4 \text{ A}$$

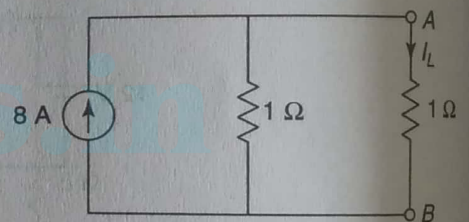
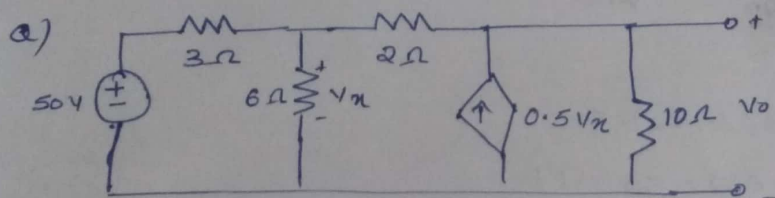


Fig. 3.357

Example 3.81

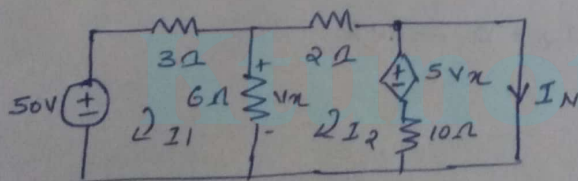
Find the current through the 1 Ω resistor.



Determine the Norton eqn ckt.

Soln

Find I_N



$$-50 + 3I_1 + 6(I_1 - I_2) = 0$$

$$9I_1 - 6I_2 = 50 \quad (1)$$

$$2I_2 + 5V_x + 10(I_2 - I_1) + 6(I_2 - I_1) = 0$$

$$2I_2 + 5[6I_1 - 6I_2] + 10I_2 + 6I_2 - 6I_1 = 10I_2$$

$$2I_2 + 30I_1 - 30I_2 + 10I_2 + 6I_2 - 6I_1 = 10I_2$$

$$24I_1 - 12I_2 = 10I_2 \quad (2)$$

$$I_1 = 11.11$$

$$I_2 = 8.33$$

$$2I_2 + 6(I_2 - I_1) = 0$$

$$8I_2 - 6I_1 = 0 \quad (3)$$

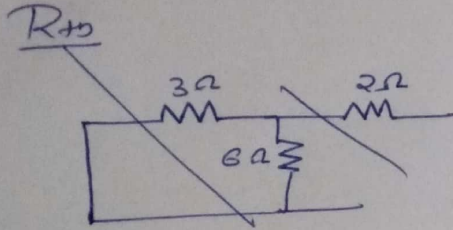
$$I_1 = 11.11$$

$$I_2 = 8.33$$

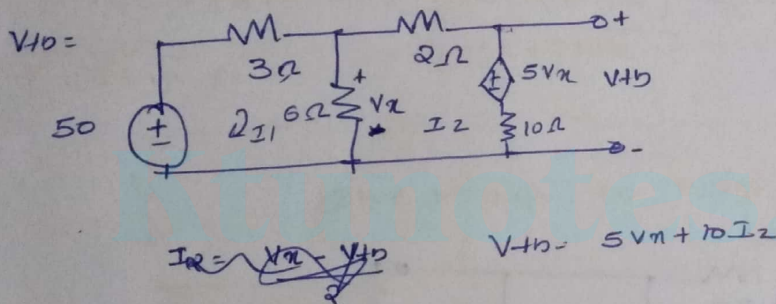
$$10I_N = 24I_1 - 12I_2$$

$$I_N = \frac{24I_1 - 12I_2}{10} = \frac{24 \times 11.11 - 12 \times 8.33}{10}$$

$$I_N = \underline{\underline{16.67}}$$



$$R_{th} = \frac{V_{th}}{I_N}$$



$$-50 + 3I_1 + 6(I_1 - I_2) = 0$$

$$9I_1 - 6I_2 = 50 \quad (1)$$

$$2I_2 + 5V_x + 10I_2 + 6(I_2 - I_1) = 0$$

$$2I_2 + 5(6I_1 - 6I_2) + 10I_2 + 6I_2 - 6I_1 = 0$$

$$2I_2 + 30I_1 - 30I_2 + 10I_2 + 6I_2 - 6I_1 = 0$$

$$24I_1 - 12I_2 = 0 \quad (2)$$

$$I_1 = -16.66$$

$$I_2 = -33.33$$

$$V_{th} = 5V_x + 10I_2$$

$$= 5(6I_1 - 6I_2) + 10I_2$$

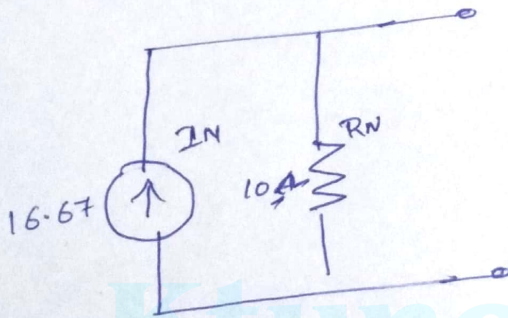
$$= 30I_1 - 30I_2 + 10I_2$$

$$= 30I_1 - 20I_2$$

$$V_{th} = \underline{\underline{166.8}}$$

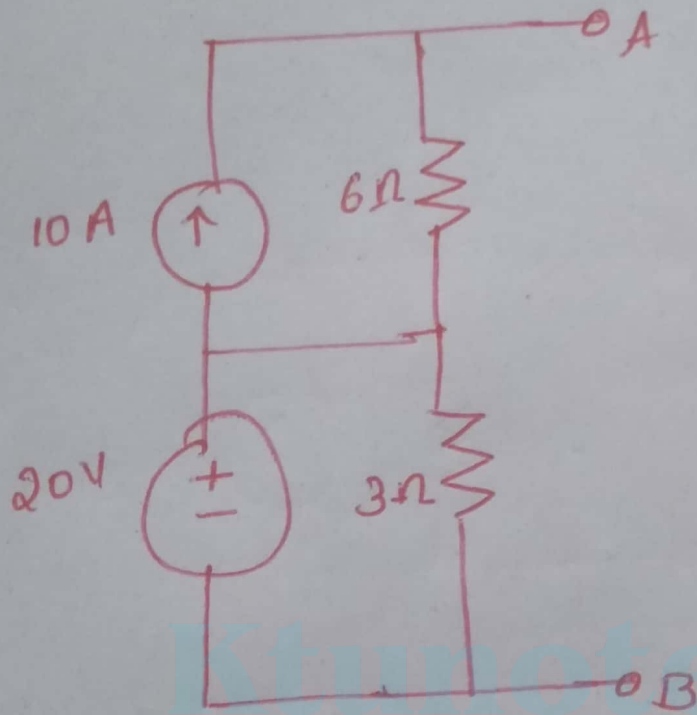
$$R_{th} = \frac{V_{th}}{I_N} = \frac{166.8}{16.67} = \underline{\underline{10 \Omega}}$$

Norton's Eqn Ckt

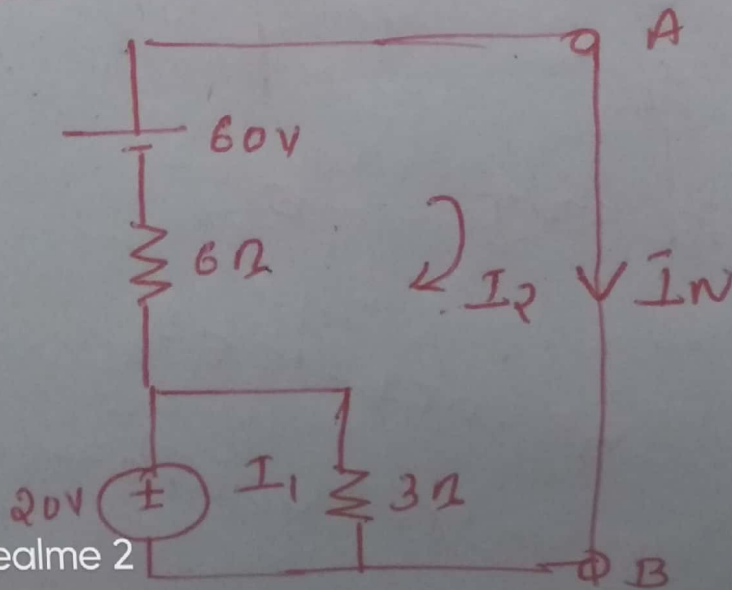


Ktunotes.in

Find the Norton Eqn ckt ?



Soln



not on realme 2

$$I_2 = I_N$$

$$-20 + 3(I_1 - I_2) = 0$$

$$3I_1 - 3I_2 = 20 \quad \text{--- (1)}$$

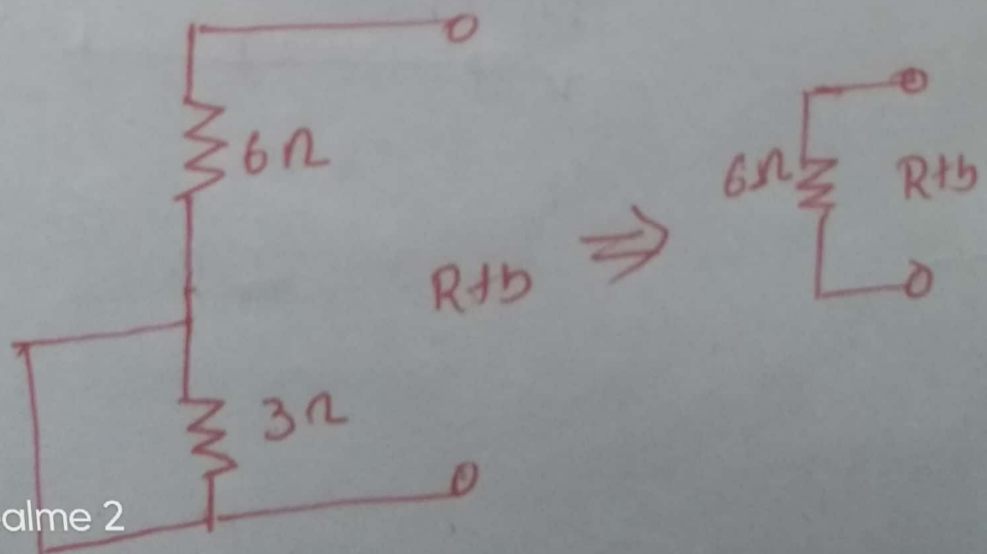
$$-60 + 3(I_2 - I_1) + 6I_2 = 0$$

$$-3I_1 + 9I_2 = 60 \quad \text{--- (2)}$$

$$I_1 = 20, \quad I_2 = 13.33$$

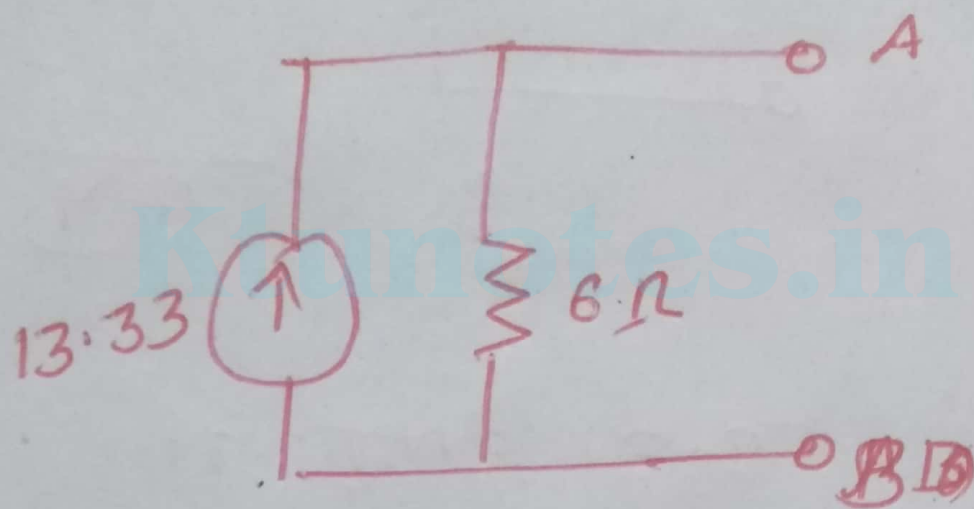
$$I_N = I_2 = 13.33$$

R_{th}



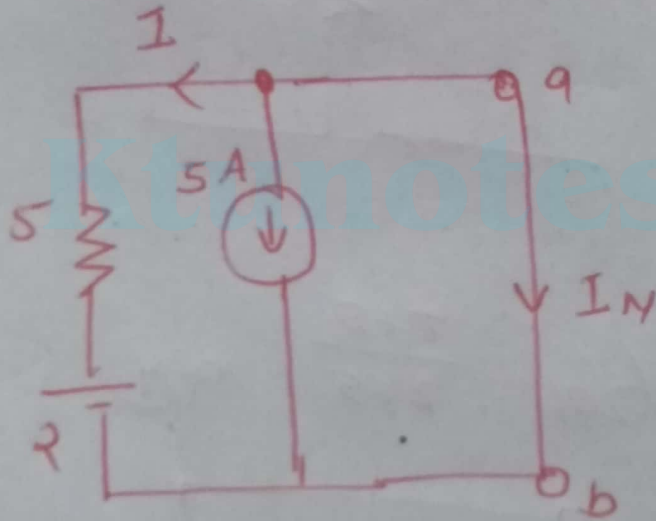
R

Shot on realme 2





Soln 1



$$I + 5 + I_N = 0$$

$$I_N = -5 - I$$

$$= -5 - \frac{V - 2}{5}$$

$$V = 0$$

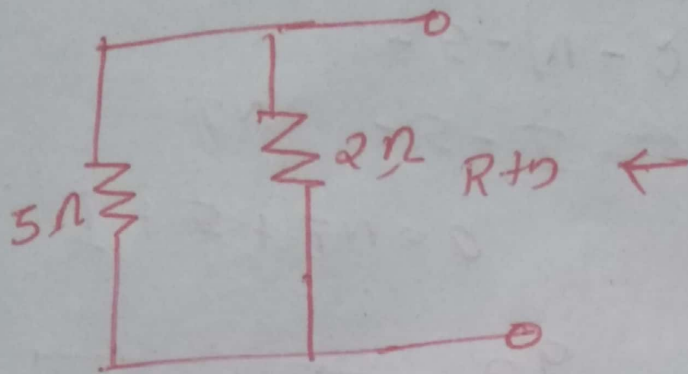
$$I_N = -5 + \frac{2}{5} = -\frac{23}{5}$$

$$= -4.6 \text{ A}$$

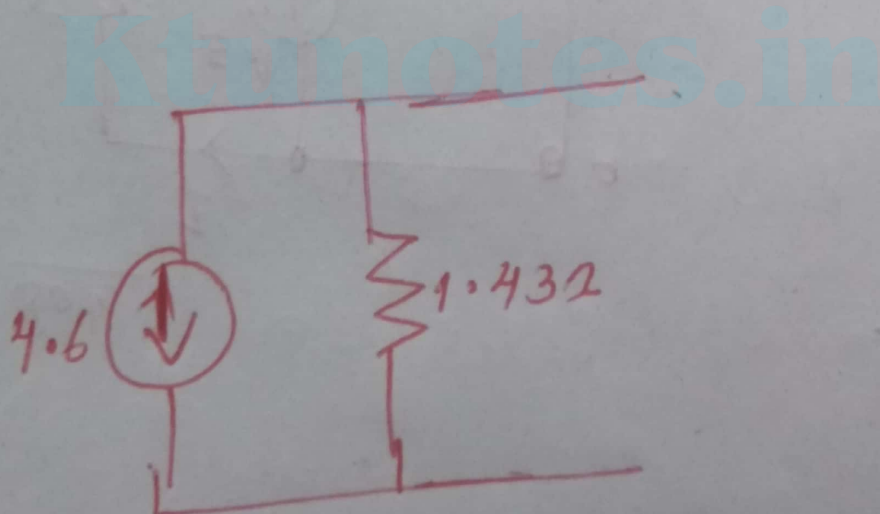
R

Shot on realme 2

R_{th}

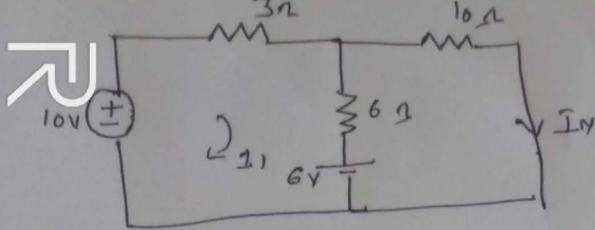


$$R_{th} = \frac{5 \times 2}{5 + 2} = \frac{10}{7} = 1.43$$



R

Shot on realme 2



$$-10 + 3I_1 + 6(I_1 - I_N) + 6 = 0$$

$$-10 + 3I_1 + 6I_1 - 6I_N + 6 = 0$$

$$9I_1 - 6I_N = 4 \quad \text{---(1)}$$

$$10I_N - 6 - 6(I_1 - I_N) = 0$$

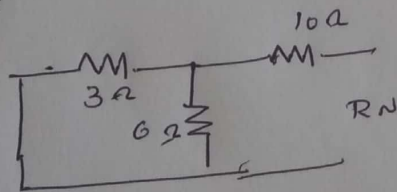
$$10I_N - 6 - 6I_1 + 6I_N = 0$$

$$-6I_1 + 16I_N = 6 \quad \text{---(2)}$$

$$\begin{bmatrix} 9 & -6 \\ -6 & 16 \end{bmatrix} \begin{bmatrix} I_1 \\ I_N \end{bmatrix} = \begin{bmatrix} 4 \\ 6 \end{bmatrix}$$

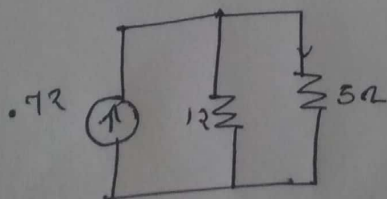
$$I_1 = 0.92 \quad I_N = \underline{\underline{0.72}}$$

$R_N =$

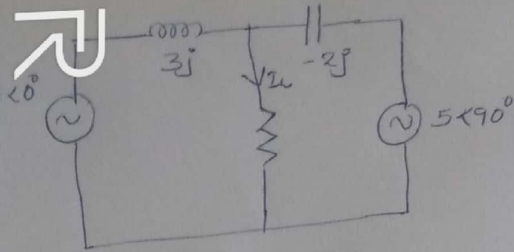


$$R_N = (3 \parallel 6) + 10$$

$$= \frac{3 \times 6}{3 + 6} + 10 = \underline{\underline{12 \Omega}}$$



$$I_L = \frac{0.72 \times 12}{12 + 5} = \underline{\underline{0.508 \text{ A}}}$$

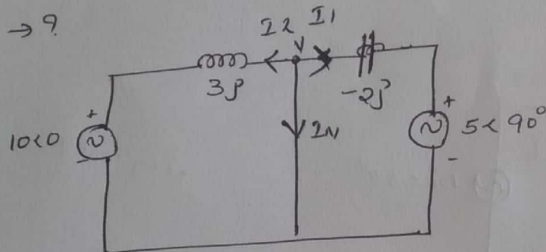


$I_{sc}(2014)$

Find the current through 5Ω resistance using Norton's Theorem?

Solu

$I_N \rightarrow ?$



$$I_1 + I_2 + I_N = 0$$

$$I_N = -(I_1 + I_2)$$

$$= - \left[\frac{V - 10\angle 0^\circ}{3j} + \frac{V - 5\angle 90^\circ}{-2j} \right]$$

Here $V = 0$

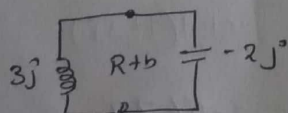
$$I_N = - \left[\frac{-10\angle 0^\circ}{3j} - \frac{5\angle 90^\circ}{-2j} \right]$$

$$= \left[\frac{10\angle 0^\circ}{3j} + \frac{5\angle 90^\circ}{-2j} \right]$$

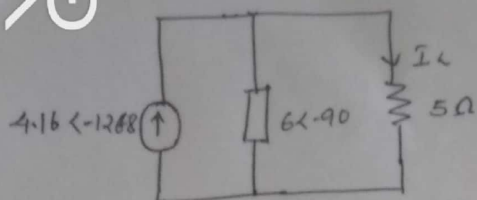
$$I_N = 3.33 \angle -90^\circ + 2.5 \angle 180^\circ$$

$$I_N = 4.16 \angle -126.8^\circ$$

R_{th}



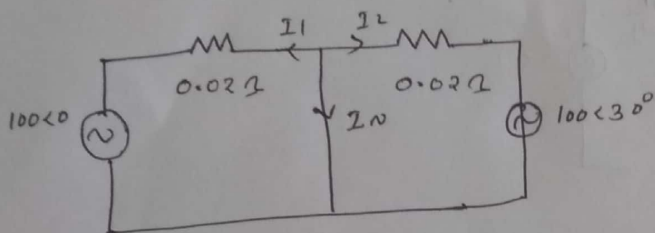
$$\begin{aligned} R_{th} &= 3j \parallel -2j = \frac{3j \times -2j}{3j + -2j} = \frac{-6j^2}{j} \\ &= -6j \\ &= 6 \angle -90^\circ \end{aligned}$$



$$I_L = \frac{I_N \times Z_N}{Z_N + Z_L} = \frac{4.16 \angle -126.8 \times 6 \angle -90}{6 \angle -90 + 5}$$

$$I_L = 3.19 \angle -166.6$$

2)



May 2019

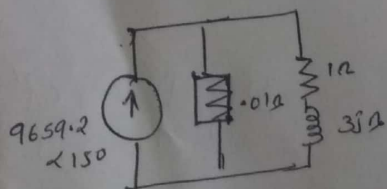
$$I_N = -(I_1 + I_2)$$

$$= -\left(\frac{V - 100 \angle 0}{0.02} + \frac{V - 100 \angle 30}{0.02} \right)$$

$$= \frac{100 \angle 0}{0.02} + \frac{100 \angle 30}{0.02} = 9659.2 \angle 15^\circ$$

$$Z_N = \Rightarrow \begin{array}{c} \circ \\ \text{0.02} \parallel Z_N \parallel \text{0.02} \\ \circ \end{array} \quad Z_N = 0.02 \parallel 0.02$$

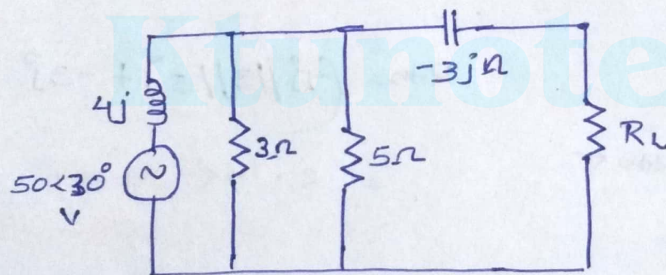
$$Z_N = \frac{0.02 \times 0.02}{0.02 + 0.02} = 0.01$$



$$I_L = \frac{9659.2 \angle 150 \times 0.01}{0.01 + (1 + 3j)}$$

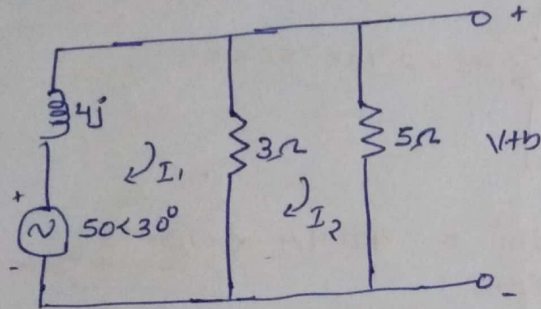
$$= 30.5 \angle -56.39$$

Q) In the network shown in fig determine the value of R_L for maximum power transfer. Also find the maximum power transferred.



Soln:

Find V_{th}



$$V_{th} = 5I_2$$

$$-50\angle 30^\circ + 4jI_1 + 3(I_1 - I_2) = 0$$

$$50\angle 30^\circ = (3 + 4j)I_1 - 3I_2 \quad (1)$$

$$5I_2 + 3(I_2 - I_1) = 0$$

$$-3I_1 + 8I_2 = 0 \quad (2)$$

$$\begin{bmatrix} 50\angle 30^\circ \\ 0 \end{bmatrix} = \begin{bmatrix} (3+4j) & -3 \\ -3 & 8 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$

$$I_2 = \frac{\begin{vmatrix} 3+4j & 50\angle 30^\circ \\ -3 & 0 \end{vmatrix}}{\begin{vmatrix} 3+4j & -3 \\ -3 & 8 \end{vmatrix}}$$

$$= \frac{0 + (3 \times [50\angle 30^\circ])}{8(3+4j) - 9}$$

$$= \frac{3 \times (50\angle 30^\circ)}{24 + 32j - 9} = \frac{3 \times (50\angle 30^\circ)}{15 + 32j}$$

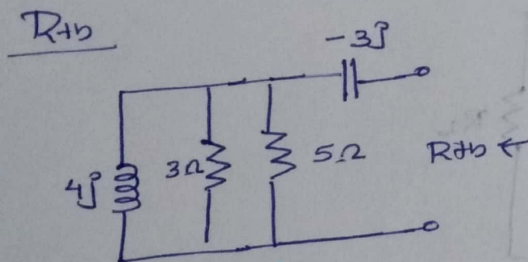
$$I_2 = 4.24 \angle -34.88^\circ$$



Shot on realme 2

$$V_{th} = 5I_2$$

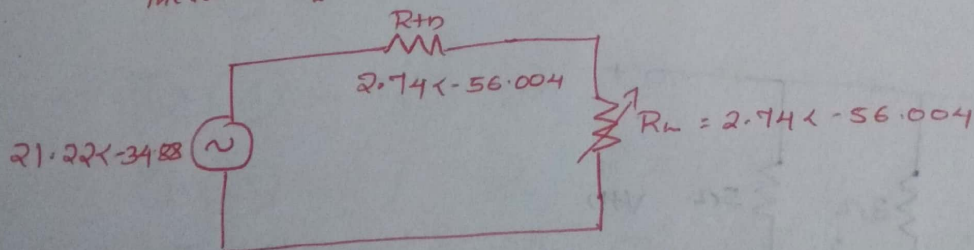
$$V_{th} = 21.22 \angle -34.88^\circ$$



$$R_{th} = (4 \parallel 3 \parallel 5) + -3$$

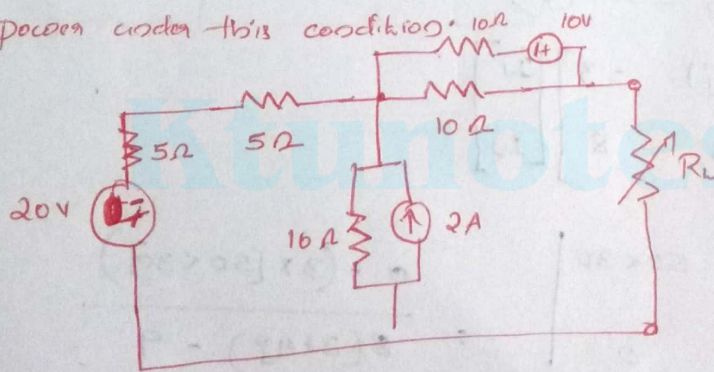
$$= 2.74 \angle -56.004^\circ$$

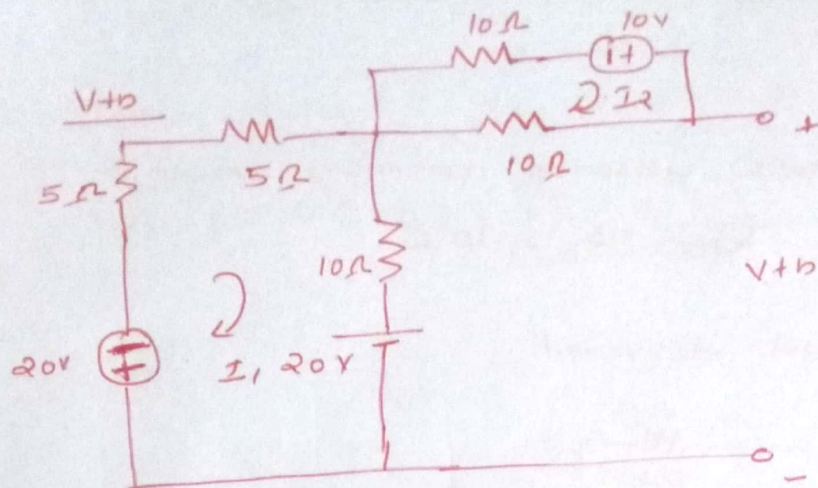
Theremin Eqn Ckk



Max Power delivered = $\frac{V_{th}^2}{4R_{th}} = 40.94 \angle -13.75^\circ$

- Q) For the circuit shown in fig, find the value of R_L that absorbs maximum power from the circuit and the corresponding power under this condition.





$$+20 + 10I_1 + 10I_1 + 20 = 0$$

$$10I_1 + 10I_1 = -40$$

$$20I_1 = -40$$

$$20I_1 = -40$$

$$I_1 = -2//$$

$$10I_2 - 10 + 10I_2 = 0$$

$$20I_2 = 10$$

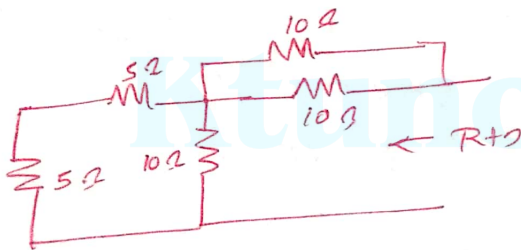
$$I_2 = \frac{1}{2} = 0.5//$$

Q

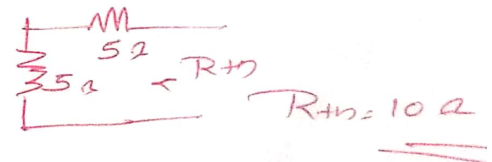
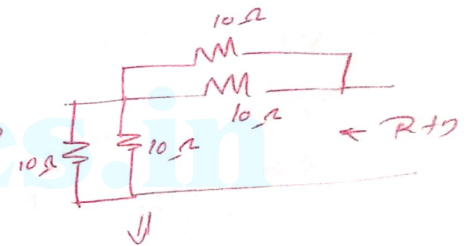
$$-10I_2 + V_{th} - 20 - 10I_1 = 0$$

$$\begin{aligned}
 V_{th} &= 10I_1 + 10I_2 + 20 \\
 &= 10 \times 2 + 10 \times 5 + 20 \\
 &= -20 + 5 + 20 \\
 V_{th} &= \underline{\underline{5 \text{ V}}}
 \end{aligned}$$

R_{th}

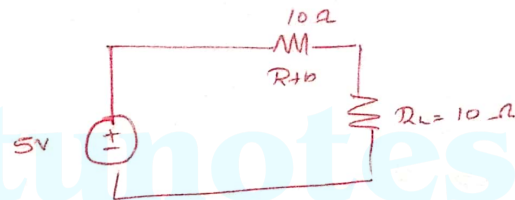


$\Rightarrow$



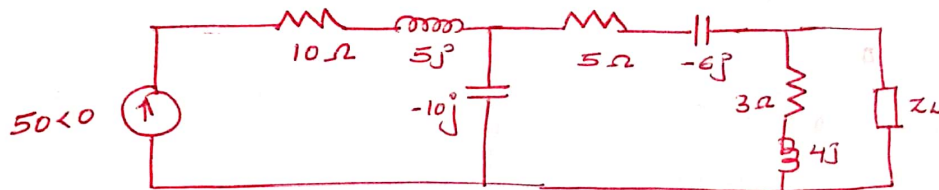
$$R_{th} = R_L = 10 \Omega$$

Maximum power delivered,



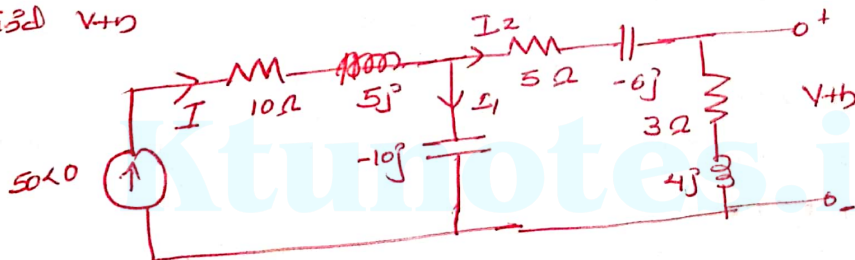
$$\text{Max Power} = \frac{V_{th}^2}{4R_{th}} = \frac{5^2}{4 \times 10} = \underline{\underline{0.625 \text{ W}}}$$

- Q) Determine the value of Z_L in the circuit shown in figure so that power delivered to the load is maximum



Soln

Find V_{th}



$$V_{th} = (3 + 4j) I_2$$

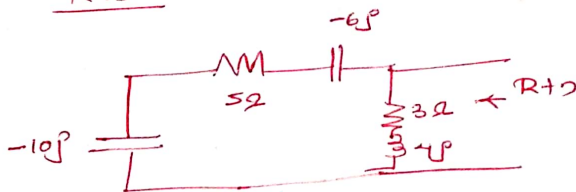
$$I_2 = \frac{I \times -10j}{-10j + 5 - 6j + 3 + 4j} = \frac{50 \angle 0^\circ \times -10j}{8 - 12j}$$

$$I_2 = 34.66 \angle -33.69^\circ$$

$$V_{th} = (3 + 4j)(34.66 \angle -33.69^\circ)$$

$$V_{th} = 173.34 \angle 19.44^\circ$$

R_{th}



$$(5 - 6j + -10j) // (3 + 4j) = R_{th}$$

$$(5 - 16j) // (3 + 4j) = R_{th}$$

$$R_{th} = \frac{(5 - 16j) \times (3 + 4j)}{5 - 16j + 3 + 4j} = 5.81 \angle 36.79^\circ$$

$$\text{or } 4.65 + 3.48j$$

$$Z_L = 5.81 \angle 36.79^\circ \text{ or } 4.65 + 3.48j$$